
MOC Operations Concept

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Attachments

Attachment Filename	Description

1. DATA ITEM DESCRIPTION

1.1. Use

Describes the proposed LMOC in terms of the operator and user needs it will fulfill, its relationship to existing systems or procedures, the satellite operational needs, and the external operational interfaces.

1.2. DID Preparation Information

The MOC Operations Concept shall address the total LMOC scope, including interactions with its external interfaces and incorporate, as directed by the USGS Landsat Mission Operations Project, operations concept and interface content received from GFP vendors. The LMOC Operations Concept shall include the following at a minimum:

- a. Flight Operations Team Organization and Roles and Responsibilities
- b. Describe how LMOC systems are operated in nominal and non-nominal modes of operations which includes at a minimum:
 - i. Mission planning processes.
 1. Off-Nadir Calibration
 - ii. Science data collection activity planning processes
 1. Instrument calibrations
 2. Primary science mission collection planning
 - iii. Maneuver planning processes
 1. Orbit modification maneuvers
 2. Inclination maneuvers
 3. Conjunction maneuvers
 - iv. Telemetry and commanding control
 1. Station contact execution
 2. Post-pass processing
 - v. Mission data management process
 - vi. Offline operations
 1. Trending and analysis
 2. Engineering data processing
 - vii. Anomaly response process
 1. Observatory onboard data recovery
 2. S/C events
 3. Ground events
- c. Address the operations concepts of any new multi-mission systems introduced as part of the design, with emphasis placed on changes to heritage operational practices.
- d. Convey how the LMOC operations concept fulfills the overall mission operations concept, and the larger Ground System operations concept.
- e. Address all the major phases of operations of the LMOC which includes at a minimum:
 - i. Pre-launch operations readiness
 - ii. Launch and commissioning
 - iii. Sustaining operations
- f. Include the approach for maintenance operations such as system patching, security scanning, and testing of software updates which might impact operations, conveying how ongoing operations are still supported concurrent with maintenance activities throughout the mission lifecycle
 - i. Include the overall approach for handling major LMOC software deliveries throughout the mission lifecycle

2. INTRODUCTION

This Operations Concept (OpsCon) will describe the intended uses of the Landsat Multi-Satellite Operations Center (LMOC) system, including the significant functional threads and use cases which drive design and testing. This operations concept covers the LMOC system, its interactions with the Landsat 8 and Landsat 9 observatories, interfaces to the Ground Network Element (GNE) and Data Processing and Archive Segment (DPAS) elements, and other externals such as NASA, NOAA, spacecraft vendor, and instrument vendor.

The Landsat System includes all components and capabilities (both space and ground based) that are under the developmental and operational responsibility of the Landsat NASA-USGS interagency team. For mission definition and formulation purposes, the Landsat System is defined at the highest level in terms of three parts - the Space Segment (including Landsat 8 and Landsat 9), the Ground System, and the Launch Segment (Figure 2-1).

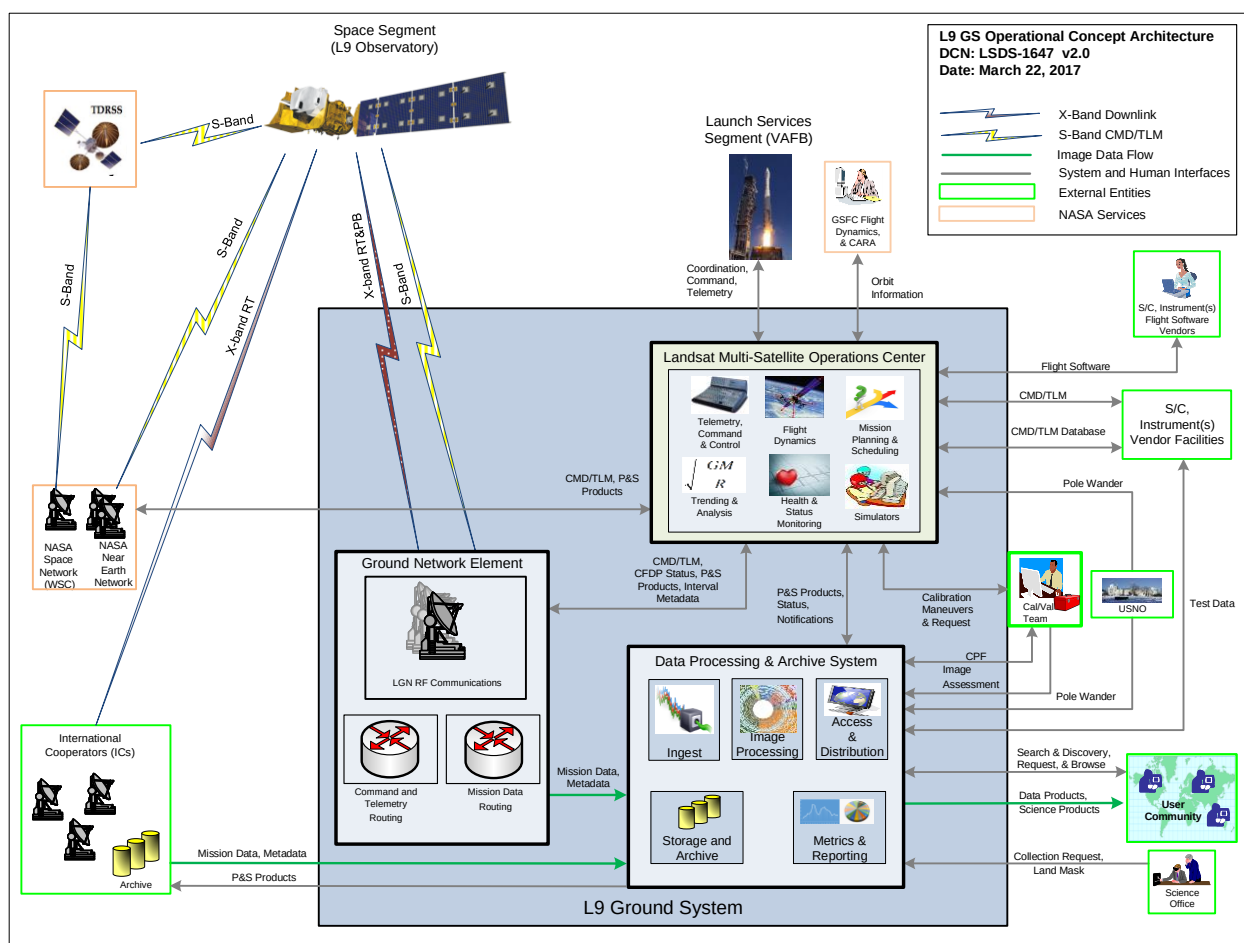


Figure 2-1 Landsat 9 Mission Architecture

The Space Segment consists of the operational Landsat 8 (L8) observatory, the Landsat 9 (L9) observatory, and pre-launch ground support equipment. Each observatory is comprised of the imaging sensor(s) and the spacecraft platform. The L8 and L9 observatories will operate in a 704 Km orbit with a 16-day repeat cycle and a 10:00 a.m. (+/- 15 minutes) mean local time for the descending node. The L8 and L9 observatories will be stationed 8 days apart within the 16

day repeat cycle. Imaging sensor and ancillary data (combined as mission data) will be collected, stored onboard and subsequently downlinked to ground stations within the ground network via an X-band communications link. This link will also include stored housekeeping telemetry. Additionally, a real-time X-band downlink capability will transmit mission (imaging sensor and ancillary) data to the Ground Network Element (GNE) / Landsat Ground Network (LGN) and International Cooperators (ICs) equipped to receive Landsat data. Each observatory will also receive and execute commands and transmit real-time housekeeping telemetry via S-band link to the ground network. L9 will utilize ground support equipment to perform ground-based integration and testing of the observatory prior to launch.

The Landsat Ground System is composed of the Ground Network Element (GNE), the Data Processing and Archive Segment (DPAS), and the LMOC. The GNE acts as the primary interface to the spacecraft and provides S-band uplink for commanding and X- and S-band downlink. The Space Network and Near Earth Network are also used in supporting satellite communication roles. DPAS processes mission data for the user community and provides input to data collection planning. The LMOC performs fleet management of the L8 and L9 observatories, ensures mission data collection and downlink, and maintains observatory health & safety.

The LMOC is based on LDCM/L8 MOC heritage and provides critical operations functions:

- Telemetry and Command
- Trending and Analysis
- Flight Dynamics
- Collection & Mission Planning
- Alert & Notification
- Infrastructure & IT Security

Key updates to the LMOC design from the heritage system include the capability to support multi-satellite operations, and replacement of the mission planning and trending subsystems. Multi-satellite configuration at the system level is largely driven by data product naming and other meta-data controls. LMOC subsystems also utilize subsystem or application level configuration controls to support multi-satellite ops and ensure intra-subsystem product and information control. Satellite-unique repository and product delivery structures both support fleet operations and LMOC system backups.

Routine operations within the LMOC will be heavily augmented by automation. Automation supports product/file transfer, pass operations, data accounting, observatory alerts, batch plot generation, and LMOC/bLMOC subsystem alerts. The operations team will monitor automation, provide sustaining engineering, plan/execute unautomated or non-routine ops functions, and provide contingency response and resolution support.

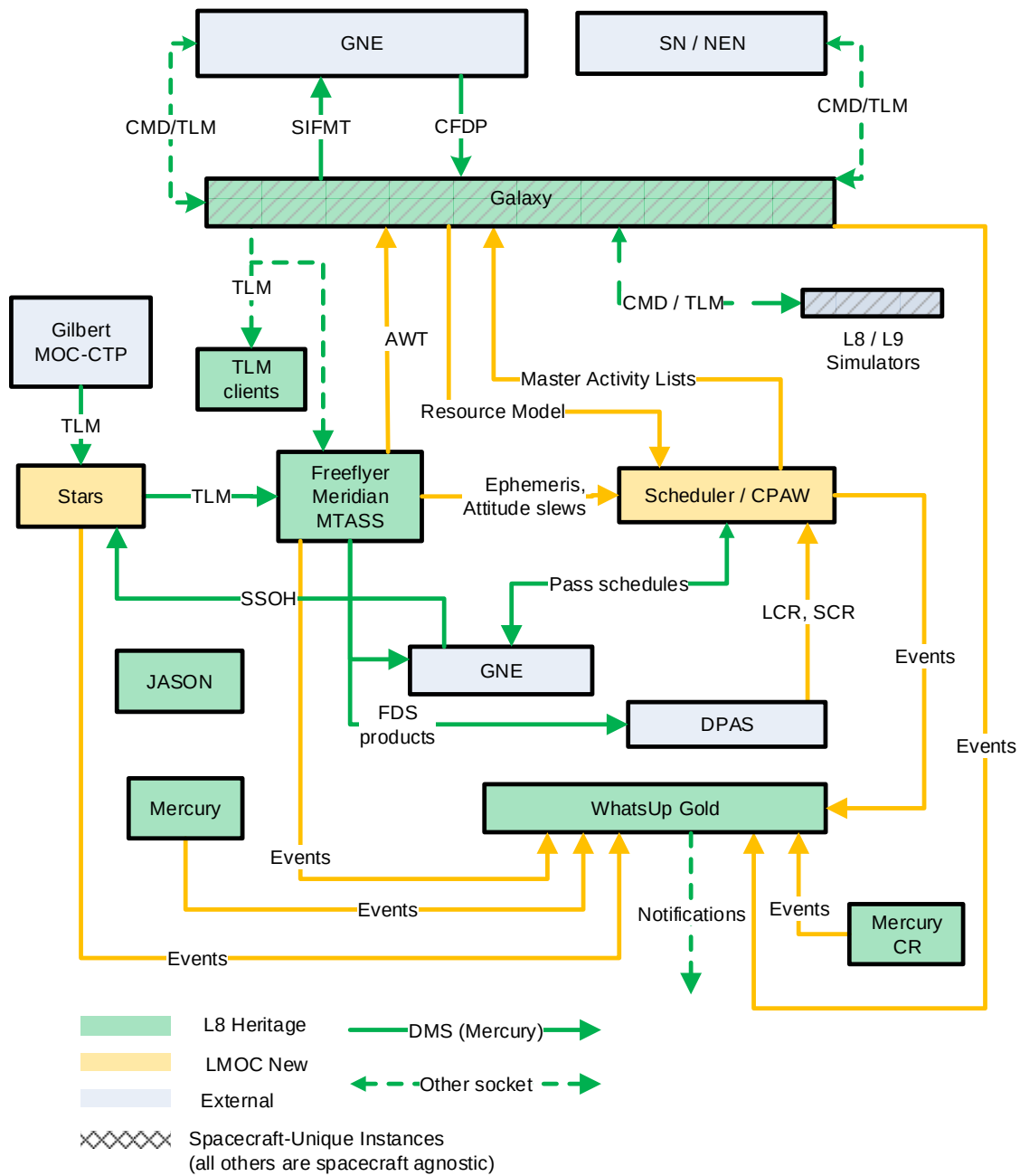


Figure 2-2 LMO Block diagram

3. FLIGHT OPERATIONS TEAM ROLES AND RESPONSIBILITIES

The LMO FOT is comprised of engineers from the GD-MS LMO Team. The team structure allows all teammates to provide unique functional support within the LMO FOT.

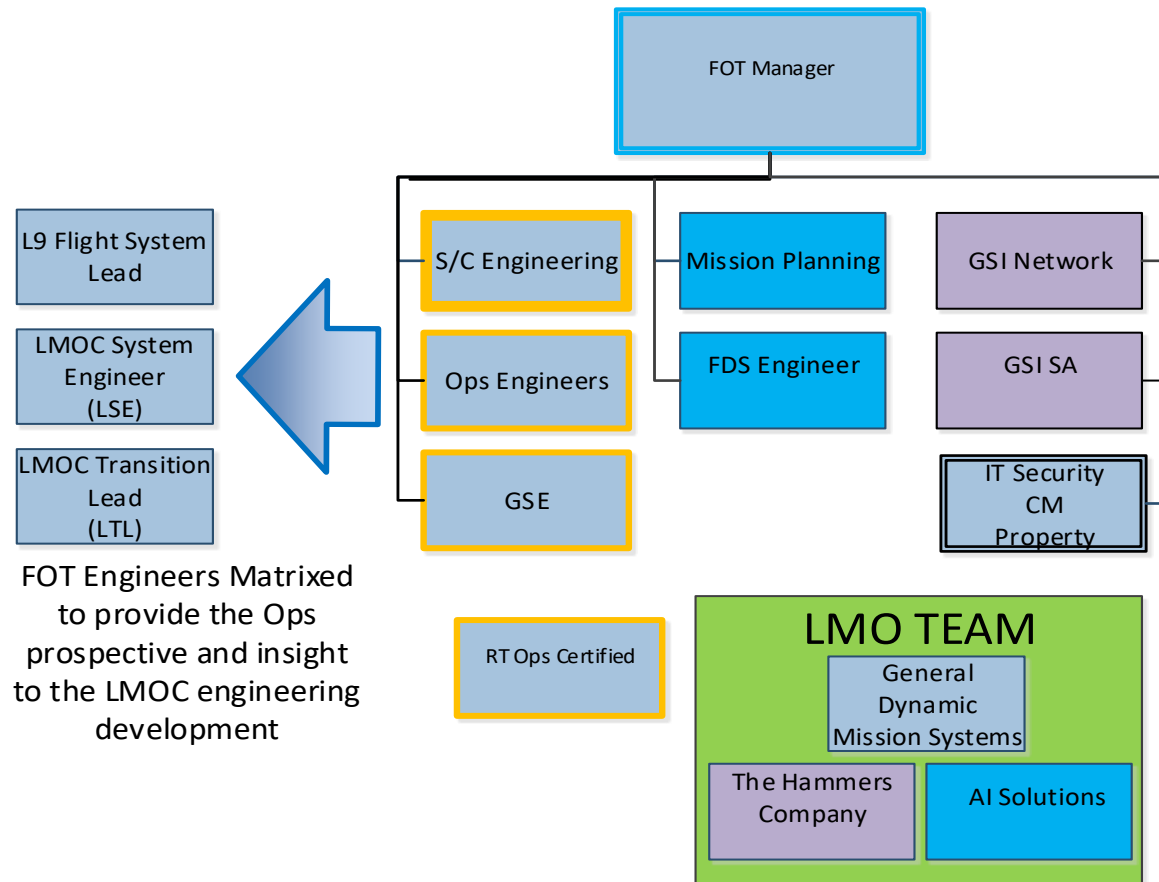


Figure 3-1 LMO FOT Functional Breakdown

3.1. Flight Operations Manager (FOM)

The FOM manages the entire Operational scope of the LMO contract and is the direct interface for the USGS Flight System Managers (FSMs). They are the day to day leadership of the FOT and work with the Flight System Lead (FSL) to ensure the resources, both internal and external, are available for seamless operation of L8 and L9. The FOM ensures that all CM best practices and processes are followed and heads the Configuration Control Board (CCB). They provide technical oversight and mentoring to the FSL. The FOM manages all staffing schedules and ensures the required training is maintained.

- The Flight Operations Manager (FOM) uses short, daily standup meetings to engage the team in clear communication of priorities, challenges, and solutions for achieving the science goals. The FOM has technical authority over the operations team.
- The FOM's major duties include leading all LMO mission operations activities and managing the FOT.

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- The FOM performs technical planning, cost and risk analyses, provides an operations perspective to systems engineering and spacecraft I&T activities, synthesizes solutions to meet USGS expectations and technical performance while staying within schedule and cost constraints, leads operations modeling and simulation activities, and chooses the appropriate standards, processes, procedures, and tools for mission operations.

3.2. S/C Engineers

S/C Engineers are responsible for the health and safety of the L8 and L9 Observatories. The L8 and L9 Flight System Leads (FSL) are the Engineers in charge of their respective observatory engineering duties. The FSLs are supported by Subsystem Engineers (SSEs) who are responsible for the various spacecraft subsystems and instruments and their components in the performance of their daily operations.

3.2.1. Flight System Lead

The FSLs are the SMEs for the L8 and L9 Observatories. They champion Observatory health and safety against scheduling and operational pressures.

- They lead the subsystem engineers in ensuring the safety of the Observatories and sustaining their performance to maximize science data collection.
- They are the FOT final technical approval for command requests before being taken to the USGS Flight System Manager (FSM) for final approval.
- They analyze trends across subsystems for interactions and Observatory-level concerns.
- They participate in all technical meetings and project status briefings as needed and lead Command Authorization Meetings (CAMs) when needed.
- They lead the FOT in anomaly/issue investigations and recoveries.

3.2.2. Subsystem Engineers

The SSEs maintain the health and safety of Observatory systems. They verify any related routine commanding, analyze trends to assess near and long term performance, and verify nominal ops for the systems. They work side by side with the Ops Engineers for special real-time command operations. (e.g. MDM, charge/discharge cycles and maneuvers).

- SSEs prepare and submit products for commanding requests and perform verification/validation activities (simulations).
- SSEs maintain the validity of procedures, displays, and configuration files for their subsystems.
- SSEs work with the instrument vendors, CalVal Team, and the Data Acquisition Manager (DAM) to prepare command requests and verify onboard execution.

3.2.3. Operations Engineers

The Operations Engineers (OEs) are tasked with day-to-day spacecraft and ground system console operations. The Ops Engineers are responsible for:

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- All normal and special S/C real-time commanding, including uplink of daily command loads.
 - Real-time ops environment readiness and execution, assessment of the Observatories during contacts, and first line response to Out of Limits (OOL) violations
 - Mission Data Management (MDM) activities, including:
 - verifying the Scene-Image Mapping Table (SIMT)
 - intervening to resolve on-board file corruption
 - working with the Ground Network Element (GNE) to quickly recover from significant MDM interruptions
 - Analysis of X-band utilization time, Solid State Recorder (SSR) file count, and SSR block usage to ensure performance margins
 - Proper operation of the Galaxy real-time ground system operations and external interfaces to the GNE stations
 - Automation performance
 - Identifying and commanding manual recorder file maintenance
 - Ground system automation performance
 - Daily console log reports

The OEs are re-certified according to the Training Plan to keep skills up to date and participate in periodic proficiency contacts and failover simulations.

3.2.4. Ground System Engineers

The Ground System Engineers are the SMEs for the various ground system applications. The GSEs are responsible for:

- Functional health and safety of the LMOC systems.
- Operational processes of the LMOC.
- Troubleshooting LMOC internal and external interfaces
- LMOC sub system performance and operations
- Performance of the file management system, real-time system operation, and trending system plot generation
- Post-pass report and stored state of health receipt
- Responding and resolving (or coordinating response) to LMOC subsystem issues.

The OEs and GSEs use time outside of contacts to maintain the LMOC systems, implement approved updates, and test possible improvements. They use time outside of that to meet training and company requirements, address operations concerns, enhance skills (e.g., further training), and read technical publications (such as from USGS.gov).

3.2.5. Mission Planners (MPs)

The MPs utilize the Mission Planning Subsystem (MPS) to generate the Master Activity List (MAL) for each Observatory; resulting in the daily Absolute Time Sequence (ATS), Imaging Window Table (IWT), and associated Scene-Interval Mapping Table (SIMT), MOC Directive File (MDF), and Confirmed Contact Schedule (CCS). They use MPS for inclusion of predictions, ingest of flight dynamics products, and handling of Observatory command requests/activities. The MPs work with instrument and science teams and SSEs to properly plan special or contingency activities, including any contact scheduling needs. They shepherd the load approval process to ensure products are available when needed for uplink.

3.2.6. Flight Dynamics

FD personnel perform a specialized role within the team. Like Mission Planning this is an offline position, with mature processes and procedures.

- FDs ensure the automated daily orbit and attitude processes (definitive and predictive) meet mission requirements and all planning products are successfully generated.
- Generate plans and command data when maneuvers are required.
- Support mission planning as needed for lunar calibration slew planning, generation of necessary command/planning products, and slew validation.
- Coordinate with the calibration and validation (CalVal) team weekly via an exchange of planning products and requests.
- Analyze planned slews and verify expected attitude Euler angles and star tracker, instrument, and radiator interference, before committal and inclusion in the operational products to ensure the safety of the Observatories.

3.2.7. Ground System Infrastructure

GSI engineers are the system and network administrators who maintain the availability, reliability, and security of the LMOC/bLMOC. They maintain the centralized, CM-controlled OS templates and the support systems needed to initialize and apply security controls to individual VMs.

- GSI engineers install security patches and system software updates, maintain network and host firewalls, and administer user accounts.
- GSI engineers troubleshoot and resolve issues with network connectivity, access permissions, and system and network performance. They set up, monitor, and maintain the system backups.
- IT security GSI engineers work with NASA/USGS IT security managers and LMO system administrators to develop solutions for compliance with federal standard security practices set forth in NIST special publications, OMB memorandums, and executive orders including, but not limited to, NIST SP 800-53 revision 4.

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- The LMO IT security engineers perform routine vulnerability scans and manage mission patch cycles on the LMOC systems. They are tasked with tracking any security variances and Plans of Actions and Milestones (POA&Ms) that make up security liens against the LMOC systems. The IT security engineers are also tasked with maintaining the LMOC IT Security Compliance Matrix (03-MD-03).

3.2.8. LMOC System Engineer (LSE)

The LMOC System Engineer works closely with the LMO Chief Engineer and LMOC Development lead in the design and implementation of the LMOC. The LSE will act as the voice of operations for the LMOC development engineers as the requirements and capabilities are decomposed and the OpsCon is nested into the design. The LSE will coordinate FOT support for development activities including I&T and T&E. The LSE will provide the deployment post-test report.

4. USE CASES AND FUNCTIONAL THREADS

4.1. Introduction to Use Cases and Functional Threads

LMOC operations are captured in a series of functional threads. These scenarios provide an overview of the major use case scenarios in the Operations Concept to describe how major activities are accomplished. This set of scenarios helps in understanding how the LMOC system behaves and serves as a tool to aid in verifying the completeness of the Operations Concept. The scenarios do not portray every possible situation that may be encountered during operations, but serve as a basis for further operations planning and building test cases for system verification and regression testing.

4.2. Major use cases:

Several use case scenarios are core to the LMOC and are required to support expected operations. These major use cases must be thoroughly understood, developed, and tested to ensure successful operations.

4.2.1. Pre-launch Operations Readiness

In the lead-up to launch, the LMOC must support a wide range of preparation and testing activities. Throughout the pre-launch phase, the LMOC supports a variety of concurrent activities, with the expected cadence and specific configurations changing over time. See Figure 4-1 for an overview of LMOC configurations in use to support parallel activities. The chief pre-launch operations activities are listed in the following subsections.

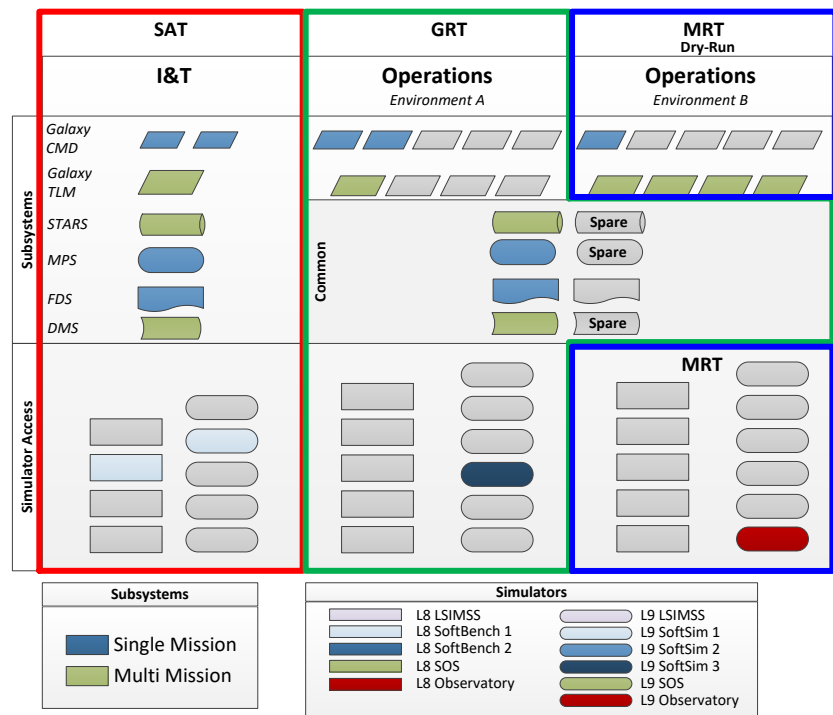


Figure 4-1 Sample Prelaunch LMOC Parallel Testing

4.2.1.1. Ground Readiness Testing, Network Readiness Testing

The purpose of Ground Readiness Tests (GRTs) is to prove the capabilities of the core integrated ground system, including the LMOC, GNE, and DPAS. These tests are organized and led by the USGS Ground System Integration & Test Team, with support from the LMO team as LMOC system users and SMEs. Network Readiness Tests (NRTs) are similar to GRTs but are geared towards testing of the LMOC with the Space Network (SN), Near Earth Network (NEN), Flight Dynamics Facility (FDF), NASA Integrated Service Network (NISN), and Conjunction Assessment Risk Analysis (CARA).

For both GRTs and NRTs, the LMOC operational environment is utilized to perform testing in a flight-like fashion. The procedures are developed to reference existing FOT Standard Operating Procedures (SOPs) to maximize reuse and to most closely mimic real operations. The specific configuration of the LMOC depends on the simulator and GRT configuration being used, with the majority of testing with the Spacecraft/Observatory Simulator (SOS) or the Linux Scalable Integrated Multimission Simulation System (LSIMSS) simulator deployed by the GNE.

The execution of GRTs and NRTs are primarily supported by the system glueware developed within the LMOC I&T environment, with additional updates as required by the FOT to address any differences between simulated interfaces and actual interfaces. The dry-run and run-for-record for these tests primarily requires a minimal number of subsystem instances for user interaction, but a performance, capstone, or LEO&C GRT may call for a larger MOC footprint.

4.2.1.2. Ops Product Development

A significant portion of the pre-launch development work undertaken by the FOT is the development of operational products which define the operation of the spacecraft and instruments. The primary types of products are

- STOL procedures (procs): Real-time scripts which perform telemetry and command operations with the observatory
- Telemetry pages, plots: Define views of telemetry used for real-time situational awareness
- SOPs & COPs: Provide FOT direction on executing pre-determined activities
- Trending plots: Define textual reports or graphical plots which render Stored State Of Health (SSOH) data captured in back-orbit and ingested by trending system
- Activity Definitions: Defines the relationship between activities planned by the MPS and actual command sequences for uplink to the spacecraft
- Mission Planning configurations: Used to generate a list of activities based on discrete input files
- Mercury model: Defines file-based product flows within the MOC
- Glueware: Other configuration files and scripting using languages such as python and bash

The LMOC supports LMOC product development by providing user access to both Configuration Managed (CM'd) and non-CM'd product spaces within user space. The end-user can utilize non-CM'd spaces for ongoing development, testing, and merging their products into the CM baseline. The LMOC also houses multiple simulators for FOT use in testing ops products, and a corresponding number of command workstations.

The LMOC also provides work management tools to organize and coordinate the work required to prepare for an MRT. The work management tools integrate to the CM system for greater situational awareness and tracking of test readiness.

4.2.1.3. Mission Readiness Testing

Mission Readiness Tests (MRTs) are formal tests conducted between the LMOC and the Observatory during the pre-launch campaign. MRTs are intended to validate ops products generated by the FOT as well as validating the ground system. The LMOC supports MRT dry-runs through rigorous CM in the operational accounts, access to high-fidelity simulators, and a multitude of smaller organizational tools. The MRT execution is supported by connecting from the LMOC T&C systems to the MOC Command & Telemetry Processor (MOC-CTP), which is provided and maintained by the GNE. The LMOC also provides other functionality as required

for MRT preparations such as FDS and MPS for load building, and Trending to assess the success of the MRT.

The NASA Mission Operations Readiness Team (MORT) leads the development and execution of the MRTs. The FOT provides input in the form of syntactically correct invocation of command procedures as requested by the MORT.

4.2.1.4. Flight Operations Team Training & Certification

The Operations Staff Training and Certification Plan (CDRL 02-OR-03) details the training activities and certifications tests required for FOT to participate in operating the spacecraft. The LMOC supports FOT training & certification through the use of simulator environments for every subsystem, and a strict segregation of data between sources. This ensures the FOT can operate the LMOC and a simulator in a high-fidelity simulation of operations. The LMOC also provides organizational tools to manage the training and certification programs.

4.2.1.5. Operational Readiness Testing

Operational Readiness Tests (ORTs) are geared towards ensuring operator proficiency and fluid operations with ground stations which have been previously tested in GRTs and/or NRTs. The ORTs are executed using only SOPs and COPs, with no other formal test procedure given.

4.2.1.6. Mission Simulations

Mission Simulations (MS's) are primarily used to train and test the wider Flight Support Team (FST) in the proper usage of the LMOC, the management of the spacecraft during LEO&C, and the coordination steps required to complete the spacecraft commissioning. The activity in an MS is centered in the Launch Support Room (LSR) which conducts all commissioning activities. The MS campaign is organized by the MORT with participation by the L9 Project, the FOT, spacecraft vendor, instrument vendor, and other support staff. The LMOC supports the MS's by providing the LSR and associated launch operations functions, and access to the high fidelity simulators. The LMO team also provides training to LSR users. The NASA project coordinates usage of the LSR and provides the MOVE voice keysets for intra-facility and inter-facility communications.

4.2.1.7. Mission Rehearsals

Mission Rehearsals are a subset of Mission Simulations which focus on the pre-launch countdown and the first few hours of commissioning. The support requirements are comparable to Mission Simulations.

4.2.2. Landsat 9 LEO&C operations

The Launch, Early Orbit, and Commissioning phase of Landsat 9 begins with the final countdown to launch and runs through the early orbit checkout of the L9 observatory.

The L9 launch countdown will begin when the NASA/KSC, launch service contractor, L9 Project, and USAF Western Range (WR) launch teams all agree that there are no impediments to launch. At T=0, the LV begins its ascent from the launch facility. During the final countdown and LV ascent telemetry monitoring and observatory support will be performed by the launch team at the launch site. During all powered flight portions of the launch and ascent through observatory separation, the LV transmits telemetry to the LV ground tracking stations and/or

instrumented aircraft. After achieving the injection orbit, the LV releases the observatory. The operations team, with support of the observatory vendors and NASA Goddard Space Flight Center (GSFC) Engineering and Technology Directorate, assumes full operations of L9 from the LMOC after the observatory separates from the LV upper stage. Following observatory release, the LV will perform a collision avoidance maneuver.

Commissioning begins at spacecraft separation from the LV. At this point the launch service contractor will provide the LMOC FDS and the NASA GSFC Flight Dynamics Facility (FDF) with a valid observatory separation state vector to support the first orbit determination. This will be used to update orbit-dependent planning products such as ground station contact data.

The spacecraft will perform an autonomous pre-programmed activation sequence after separation from the LV to initiate deployments, generate power to the observatory, and turn on critical subsystems. During early-orbit, the proper functionality of observatory command and data handling functions and attitude and orbit determination and control functions will be demonstrated. The L9 observatory will be capable of transmitting real-time S-band telemetry to the SN for routing to the LMOC T&C system. Additionally, the observatory may begin transmitting real-time S-band telemetry to the GNE ground stations and the NASA Near Earth Network. The observatory X-band system will be enabled and checked out against the GNE prior to instrument activation.

During commissioning, activation of the L9 imaging sensors begins. The imaging sensors will be activated based on a series of steps following out-gassing, including powering-on of imaging sensor electronics, functional checkout of any on-board calibration sources, generation of imaging sensor health and status, and collection of the first image. Imaging sensor data will be transmitted via X-band communications to the GNE ground stations.

The LMOC will receive telemetry through the ground network and will begin monitoring spacecraft status and health. Imaging sensor data will also be received by the GNE and routed to the DPAS.

After initial observatory checkout steps, it is anticipated that orbit adjustments will be made to move the observatory from its initial parking orbit to its nominal orbit. This will include any coordination needed to bring the observatory into the “morning constellation” of Earth-observing satellites. These orbit maneuvers may take several weeks and will require periodic orbit determination using observatory telemetry and flight dynamics analysis. The NASA GSFC FDF will be available to support these activities if needed.

It is anticipated that activities such as spacecraft checkout, orbit raising, imaging sensor(s) out-gassing, activation and decontamination will nominally take place in the first 30 days following launch. Likewise, days 30-60 will nominally include imaging sensor(s) transition to operational mode, first light image collection, and start of on-orbit imaging sensor(s) calibration activities with mission data flow tests.

During L9 LEO&C the operations team will be supporting ongoing operations of L8. Since L8 is in the routine operations mission phase it is anticipated that automation will reduce operations staffing need. The operations team will be compartmentalized between L8 and L9 operations to ensure any L8 issues do not disrupt L9 observatory commissioning. Additional details regarding simultaneous satellite operation can be referenced in the Transition Plan (MT-01).

4.2.3. Mission Transition to LMOC

The LMOC and bLMOC will be thoroughly tested prior to starting operations with the L8 and L9 observatories. LMOC and bLMOC testing will support the L9 I&T plan, which includes Mission Operations Simulations (MOSs), Mission Readiness Tests (MRTs), and Ground Readiness Tests (GRTs). The L9 MOS, MRT, and GRT plans will test LMOC/bLMOC internal functions and interfaces with the L9 mission spacecraft, GNE/ /SN/NEN, EROS, FDF, and other external segments.

Since L9 testing must be accomplished on an extremely strict timeline to meet the L9 launch date, L8 testing of the LMOC/bLMOC may not impact L9 testing. L8 testing is completely independent of the L9 MOS, MRT, and GRT plans within the LMOC/bLMOC. Successful completion of the L9 I&T plan will certify the LMOC/bLMOC for L9 operations.

The L8 transition will be handled separately from L9 and will focus on verifying L8 ground/operations functions. Note, fleet operations (L8 and L9) will be tested and validated as part of LMOC I&T work. Network Proficiency Tests (NPTs), Operation Readiness Exercises (OREs) and Transition Readiness Tests (TRTs) have been outlined to verify LMOC/bLMOC functionality, external interfaces, and ground system/ground network interactions.

NPTs are informal tests which will exercise the L8 operations interfaces. NPTs will demonstrate network connectivity and appropriate accounts/privileges required for data flows and external interfaces. NPTs will be conducted between the LMOC/bLMOC and the following external interfaces (all other external interfaces will be tested via LMOC I&T):

- SNAS
- S/OS
- SB
- Vendor Delivery
- Alert Path
- SN CMD/TLM
- NEN CMD/TLM
- LGN CMD/TLM

OREs are designed to demonstrate basic L8 operations functions on LMOC subsystems. All

OREs will utilize simulator data instead of real-time data. Several OREs will require passing test products between external interfaces and LMOC subsystems.

TRTs are formal system level tests and will use the observatory simulators to replicate day in the life operations and LMOC functionality.

- TRT Functional Threads (L8 Only)
- TRT Shadow Operations Testing (L8 only, DITL and TWITL)
- TRT Multi Satellite Testing

LMO will coordinate the tests with the other GS elements and take their schedule into account for testing.

For more details on L8 Transition, refer to CDRL_04_MT_01 L8 MOC Transition Plan.

4.2.4. Day in the life science operations (DITL)

Nominal Landsat operations revolve around the day in the life (DITL) cadence. Central to the DITL is the daily activity scheduling and execution of observatory commands. The L8 and L9 satellites execute stored command loads that contain instructions regarding spacecraft subsystem management, image collection, instrument calibration, transmitter management, and data downlink. These command loads must be built, reviewed, and uplinked on a reoccurring schedule to keep a fresh command load on each satellite for execution.

The timeline shown in Figure 4-2 outlines the cadence of routine operations contacts as well as load-building activities for the scenario where both spacecraft loads are built on the same day. This scenario assumes a single workstation is being utilized for both spacecraft, the operations team may elect to utilize two workstations in parallel for two spacecraft if the daily staffing permits.

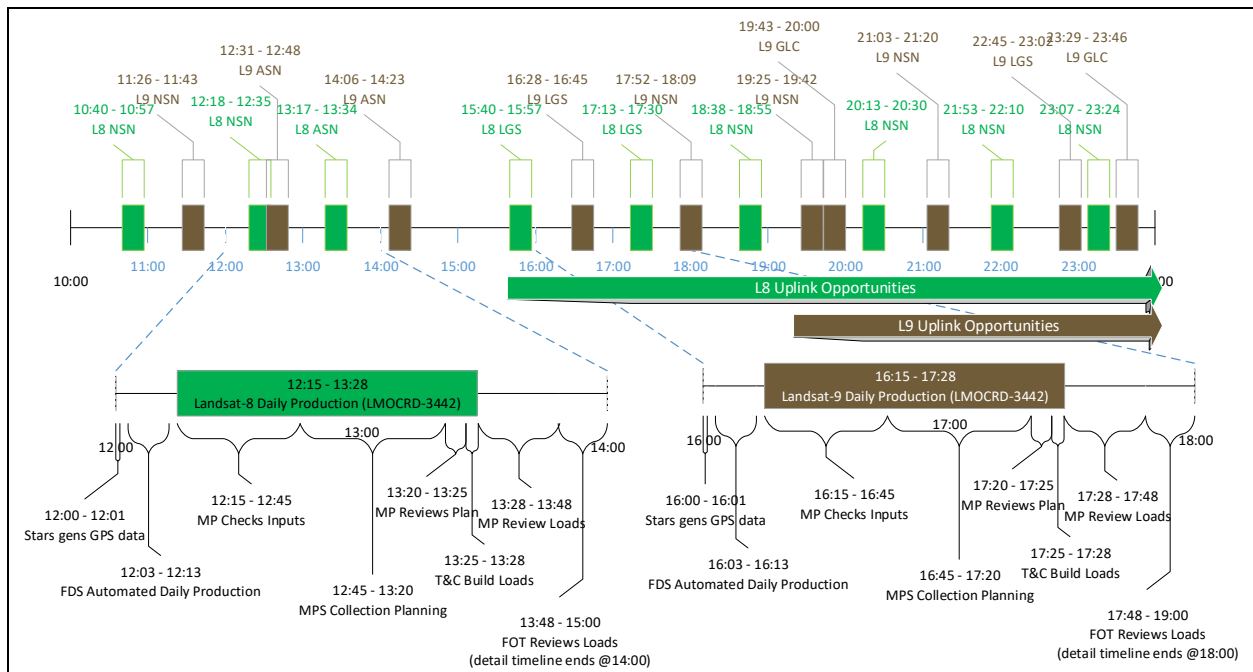


Figure 4-2 Routine Daily Operations

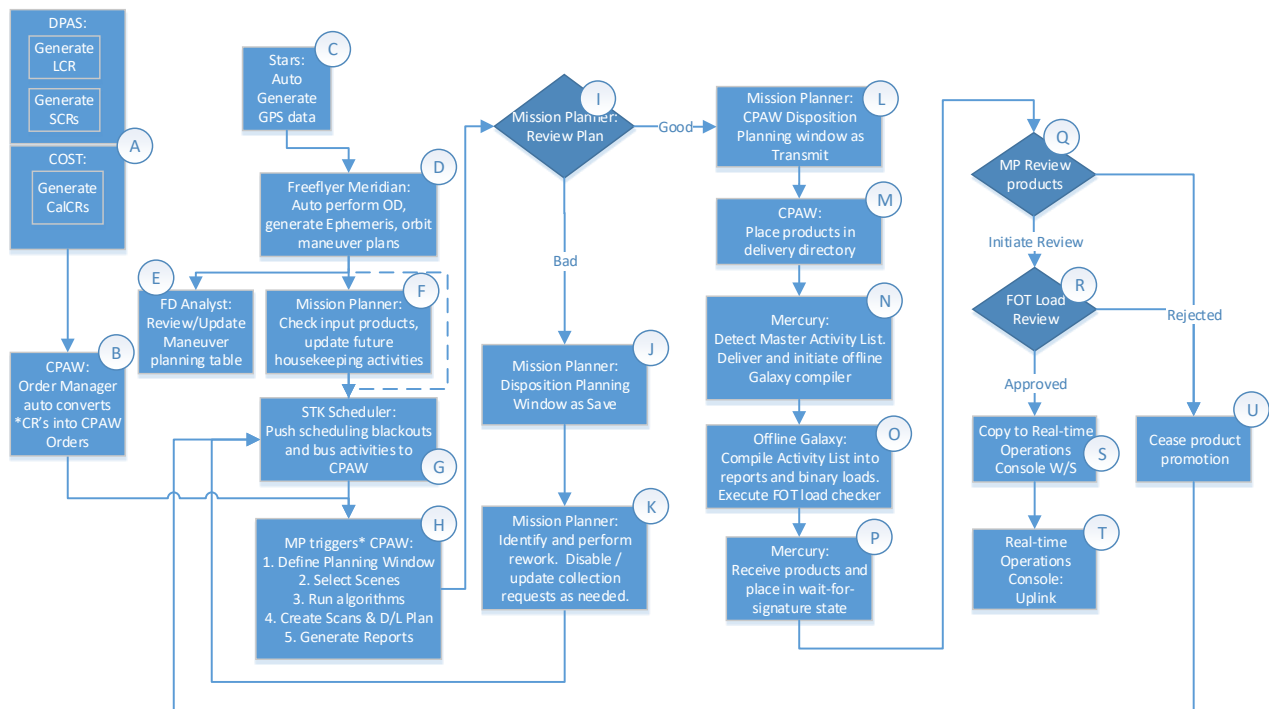


Figure 4-3 LMOC MPS DITL

Prior to beginning command load generation, several input files must be created and collected.

A. Science community user requests are reviewed by the USGS Data Acquisition Manager and approved requests are posted to DPAS as special requests. Calibration Collection Requests are generated by the LMOC-internal COST system.

- B. The Collection Planning and Analysis Workstation (CPAW) converts the collection requests into CPAW orders.
- C. STARS collects real-time and back-orbit spacecraft data and generates the GPS data that will be used by Flight Dynamics.
- D. Freeflyer Meridian uses this GPS data to create various ephemerides and orbit maneuver plans.
- E. The Mission Planner checks all of the input products and updates future housekeeping activities.
- F. Flight Dynamics reviews and update maneuver products if necessary.

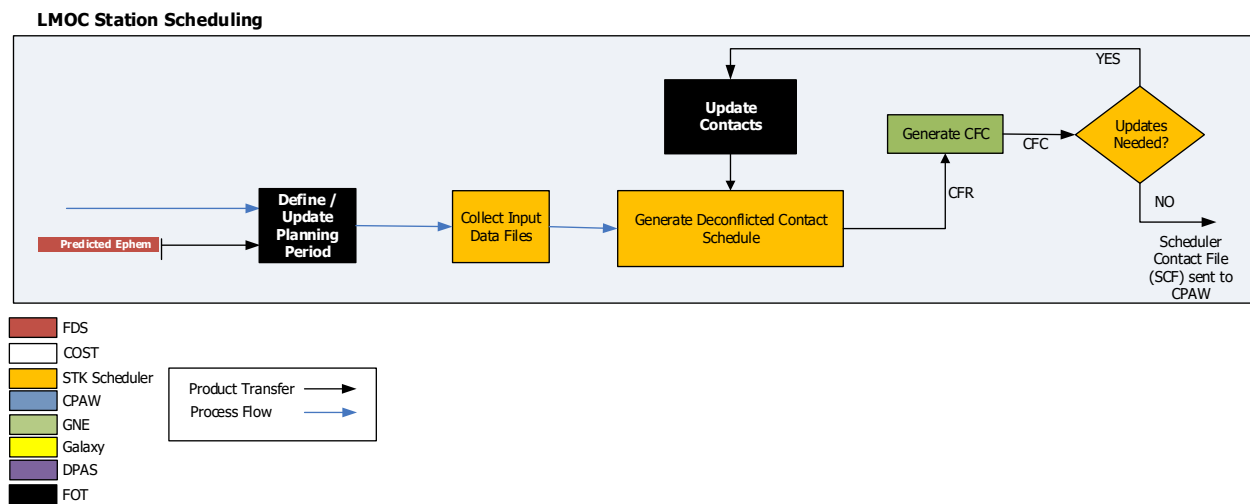


Figure 4-4 Station Scheduling

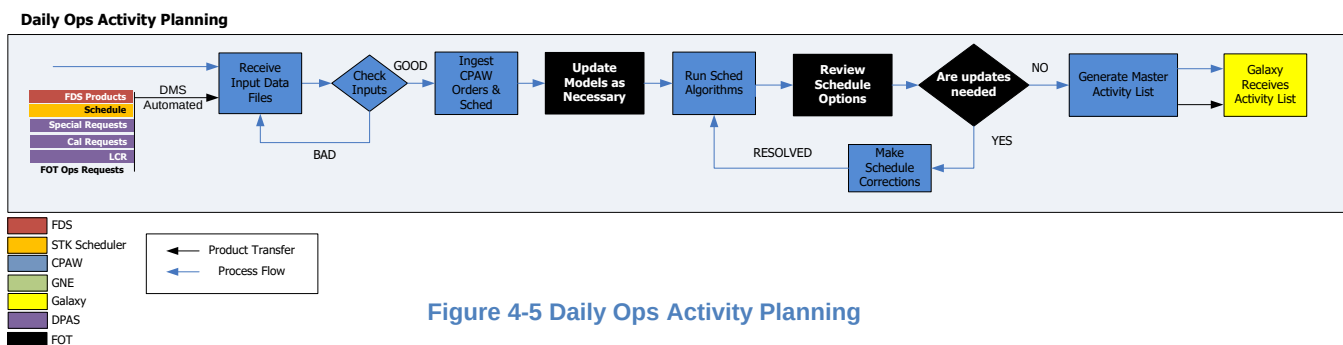


Figure 4-5 Daily Ops Activity Planning

Each week the mission planning team submits acquisition products to GNE/NEN stations with a requested contact schedule. The requested contact schedule is created in STK Scheduler using predictive ephemerides provided by FDS and scheduling rules programmed into STK Scheduler. STK Scheduler will create a Contact Forecast Request that contains the desired ground contacts for the next 28 days. Stations will review the requests and provide a Contact Forecast Confirmation with the contacts they can support. A Scheduler Contact File (SCF) is then sent from Scheduler to CPAW.

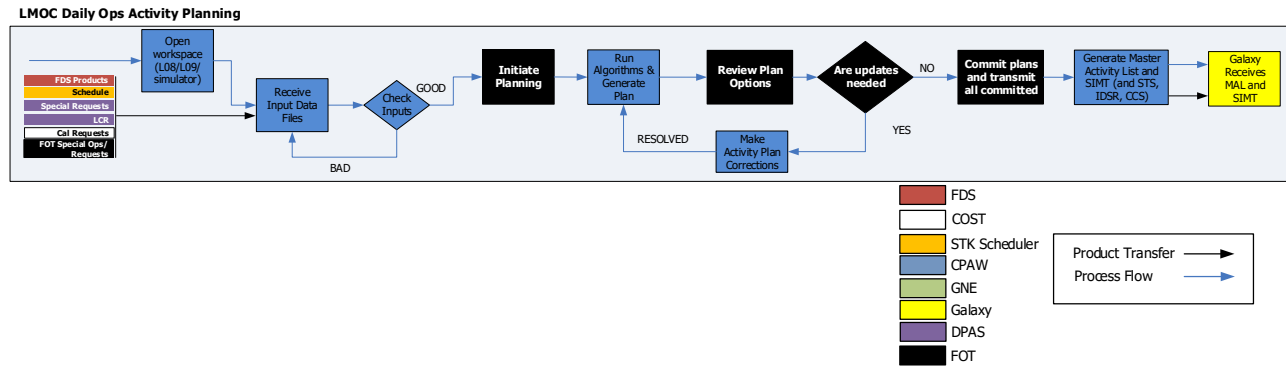


Figure 4-6 Activity Planning

- G. The Mission Planner collects current planning inputs - (Scheduler Contact File, LCR, SCRs, CalCRs, recorder state, and subsystem command requests) and creates a Master Activity List (MAL) in CPAW for each satellite. The MAL is essentially a deconflicted activity plan that serves as the basis for the satellite command load. A typical MAL will contain instrument commands for image collections or calibrations, data playback commands, and C&DH maintenance commands. The standard MAL duration for both spacecraft is 96 hours. In parallel to the MAL, the CPAW software generates the SIMT, STSs, IDSR and CCSs.
- H. The Mission Planners internally review the plan.
- I. If an issue with the plan has been identified, the planning team will “save” the plan, and
- J. Start the rework and recycle to step G.
- K. If no issues are identified, the planning team marks the plan as “transmit” and
- L. Delivers the products to the “Delivery” directory.
- M. Mercury will detect that the MAL has been delivered and start the offline Galaxy compile.
- N. Galaxy uses the Stored Command Sequence (SCS) compiler to reformat the MAL for each satellite into binary load by using the Galaxy command database to convert instructions. The SCS also generates text versions of the command loads for operations review and checks. Text versions of the command load are then used as inputs to the operations load checking script. The load checker examines the load for configurations, values, and event conflicts which should be avoided.
- O. The outputs from SCS (binaries and text versions) and the load checker for each individual satellite are grouped together and collected by Mercury for operations review and signature.

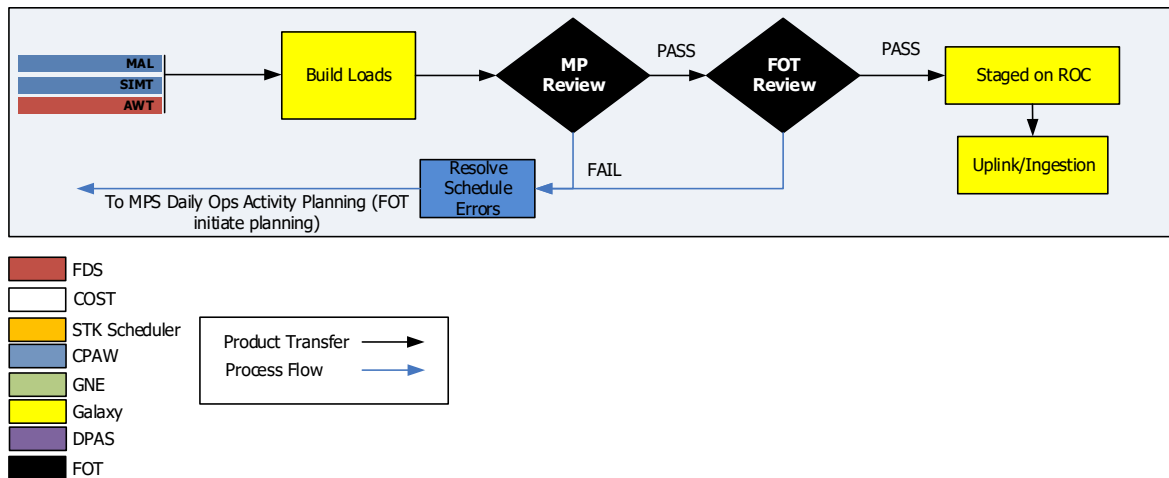


Figure 4-7 Load Generation

- P. The Mission Planning team does a final review and if approved, forwards to the FOT for review. If the load is rejected, the process cycles back to step G.
- Q. The FOT reviews and if approved...
- R. The FOT copies the load to the Real-time Operations Console (ROC). If they reject the load, the process cycles back to step G. Typical required review and signatories include a Spacecraft Engineer and an Operations Engineer.
- S. Ideally, a command load is delivered to a Real-time Operations Console 3 uplink passes before UTC day change. Depending on the time of year this would be approximately 2 p.m. or 3 p.m. local time. The Operations Engineer either manually, or the ROC autonomously uplinks the load to the S/C.

Command loads that have been uplinked to the satellite will sit in an inactive buffer in FSW until UTC day change. At this point the active and inactive command buffers will swap and the new command products will become active. As mentioned earlier, command products contain a minimum of 72 hours of commands, so a new command load is not required to be uplinked daily.

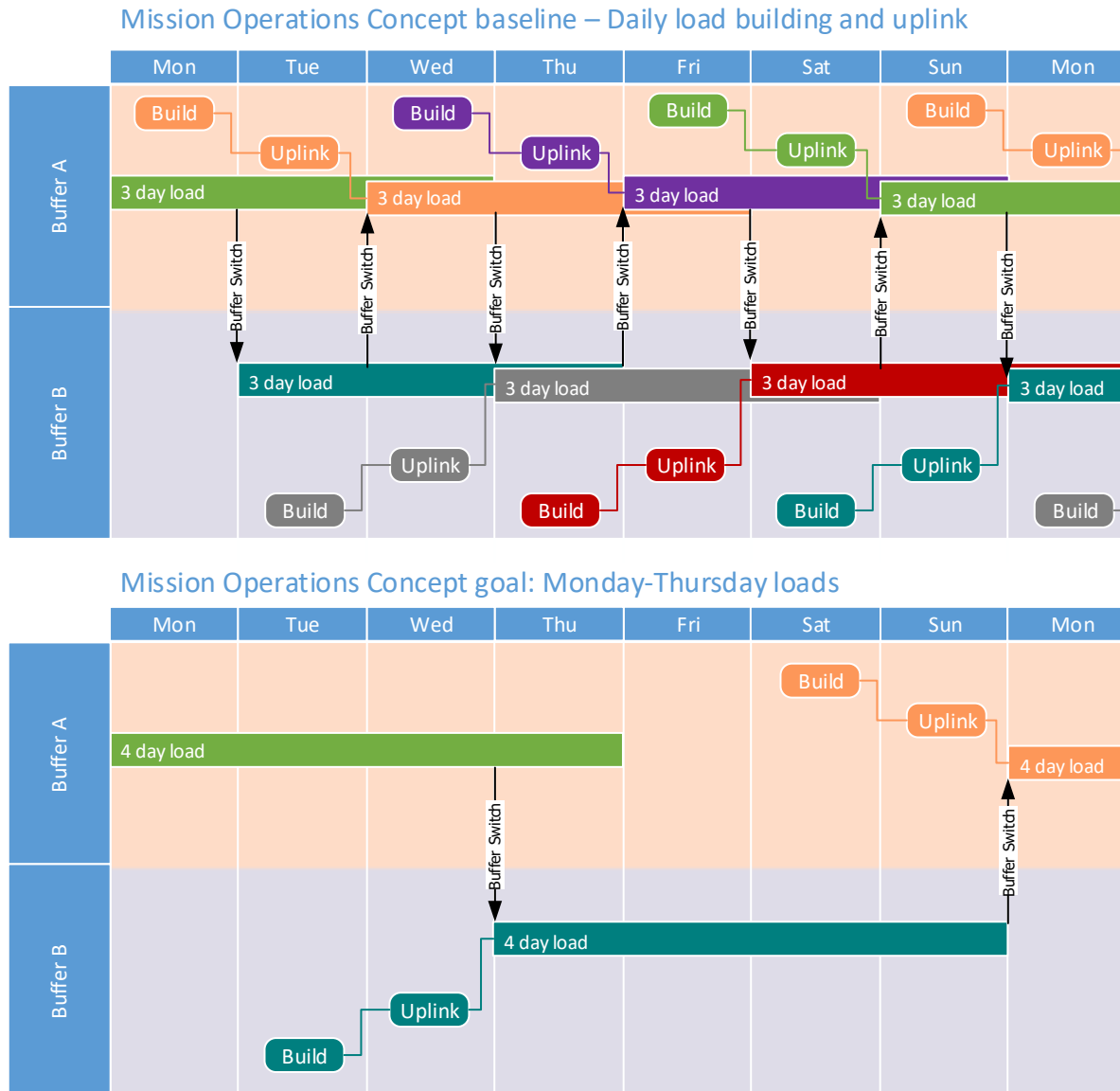


Figure 4-8 Load Building, Uplink and Operation

Figure 4-8 shows the baseline plan for LMOC load building, uplink and operation on spacecraft buffers (top), and the eventual goal of building, uplinking and operating loads twice a week (bottom). The baseline plan is driven from L8 Mission Planning heritage, whereby a load is built on day 1, uplinked on day 2, and run on day 3. In this baseline, loads are built and uplinked every day and a buffer switch also occurs every day at approximately 00:00 UTC, to a stored command load on the other buffer. The Mission Operations Concept goal reduces the amount of loads built to two four-day loads per week, per spacecraft. A stored command margin of one day will always be preserved onboard.

Typically, each satellite will take one GNE contact per 90 minute orbit. Since the L8 and L9 satellite orbits are 180 degrees out of phase they will never be in contact with the same ground station at the same time after L9 exits the LEO&C phase. It is possible, however, for each satellite to be in contact with the ground at the same time on different ground stations. Each satellite will have a separate Real-time Operations Console (ROC) with a schedule executer and file accounting instance to ensure a designated command path and pass automation is available for each satellite at all times.

Each satellite ROC will run automation that will configure for and execute nominal GNE NEN pass operations. A unique MDF is delivered to each satellite ROC which contains the pass times scheduled as part of the daily load processes associated with that satellite. At day change the ROC schedule executer activates the newest MDF for cueing pass operations. The MDF contains the AOS and LOS times for pass times associated with the corresponding command load. Schedule Executer on the ROC uses the AOS and LOS times within the MDF to kick off ground automation procs used for ROC pre-pass configuration, automated real-time commanding, and ROC post-pass configuration teardown.

Prior to a scheduled ground contact on the GNE/NEN, the MDF and Schedule Executer will kick off ground automation procs which configure and establish links to the ground station, configure the command encryption hardware, and open a new log. After AOS ground automation will issue a NOOP command to the satellite to verify command link, then proceed with issuing retransmits and performing commanding/product uplink required for MDM. During the pass, the active command load on the satellite will play back science data and back-orbit housekeeping data. After LOS the links to the station are terminated, the GALAXY log is closed, and a post-pass report is created.

Pass automation is nominally active on both the L8 and L9 ROCs at all times, but is monitored closely by the Operations Engineer (OE). The OEs are responsible for monitoring automation and resolving any noted issues with ROC configuration, links to GNE/NEN/SN stations, and automation. Occasionally commands that are not part of nominal automation must be issued. Command procs which require execution are submitted via a Command Request (CR) and must obtain approval signatures. Once CR approval is obtained the OE will coordinate with the requestor and execute the desired command procedure at a coordinated time.

A bank of Galaxy workstations will be located in the LMOC for spacecraft engineering use. When in contact with an observatory, telemetry can be forwarded from the ROC to Galaxy workstations and other subsystems. Galaxy will be able to decom telemetry from either L8 or L9 spacecraft, and the user can acquire the telemetry stream that they choose. To prevent operator confusion, separate telemetry pages and careful display design will be used to differentiate satellites.

Several other LMOC systems incorporate automation to ease the operations workload and to provide quality or configuration checks. At the heart of this distributed automation is the Mercury subsystem, which distributes products between systems within the LMOC and to/from

external interfaces. The Mercury system monitors several subsystem directories on the NFS within the LMOC for files which match a programmed set of conditions. Such conditions (file name, size [requires additional scripting], source, destinations, grouping, etc.) are documented within the Mercury model and control movement and manipulation of files. A signature condition can also be specified as part of the model. File movement can be blocked until required signatures have been electronically provided through the Mercury interface. When required signatures have been provided, the file will be delivered to the intended destination. File signatures act as a way to ensure operationally sensitive files have been reviewed and checked. Since Mercury is designed to be a generic data management tool, the Mercury instance within the LMOC will handle the movement of both L8 and L9 files. Important operational files are also copied to the Controlled Repository (CR) when they are identified by Mercury. The CR serves as a central location where operations products can be stored for operations team search/retrieval and archive. The CR has an LMOC internal web interface which the operations team can use to find and retrieve important operations products for review or analysis. External entities send and retrieve files to the Controlled Repository (CR). The CR is an outward facing file server which is used as a file exchange server with the outside world, this is accomplished with the use of the Junos Firewall.

The STARS subsystem also uses decentralized automation to produce plots and data files for other subsystems and the operations team. After the end of a nominal ground station pass, the ground station will send the stored state of health (SSOH) data to EROS, which in turn sends the data to the LMOC. SSOH data contains the back-orbit housekeeping and engineering data produced by the satellite while out of contact. Once a SSOH file is delivered to the LMOC, the STARS system checks the file time range against the expected time range as indicated by the stored command loads that generated the file. If the SSOH file time range looks correct, it is then checked for data volume. SSOH files with the correct data volume are processed, whereas SSOH files which are missing data or do not cover the expected time range are marked for retransmit from the satellite. Good SSOH files are injected into STARS and stored in the trending database. During the data injection process, several data products are produced for spacecraft engineers (limit violations), operations (requested sequential prints), and Flight Dynamics (post-pass and DTM GPS and attitude products). As the data is injected into STARS, data statistics are calculated and stored within the STARS database. Since the STARS database can be binned by source (L8 spacecraft, L9 spacecraft, Simulator, etc.), the LMOC will have one STARS database for storage of data for all mission spacecraft and simulators. Once per day separate system automation will kick off to produce engineering batch plots for spacecraft engineer analysis and spacecraft attitude and position products for use by Flight Dynamics. Users can submit ad hoc data or plot requests to STARS through an API or web browser interface.

Another important piece of LMOC automation is the Alerting and Notification Subsystem (ANS) which provides the operations team alert notifications. The ANS provides monitoring and alerts in the following main areas: spacecraft status, real-time operations/automation status, and ground subsystem/automation status. Spacecraft status is monitored by the ANS receiving and parsing log files from the ROC and limit violation reports from STARS. Identified misconfigurations, limit violations, or watchdog failures will be sent to the appropriate operations team members.

The ANS also receives logs from all of the major LMOC subsystems (ROC, FDS, STARS, Mercury, Galaxy). Logs are parsed and checked for subsystem errors, timeouts, or misconfigurations that require ops team alert. The ANS will distribute alerts to the appropriate notification list via email and SMS. As part of receiving an alert, the operations team will reply to ANS to notify ANS that the alert was received. The ANS system is intended to be active at all times (whether the LMOC is manned or unmanned) and handles alerts for both the L8 and L9 satellite.

Operations is supported by several ancillary software tools to aid in documentation, workflow tracking, and product development. General notes, operator logs, and miscellaneous information will be tracked in the LMOC wiki. While there will be one prime Wiki application in the LMOC, the Wiki can be organized into separate L8, L9, and LMOC subsystem entries. Operations and system administration tasks will be organized and tracked within Jira. Similarly to the Wiki, there will be one prime Jira instance for the LMOC, which will be organized for L8, L9, LMOC subsystems, and administration tasks. Distinct workflows will be defined for each task type to ensure appropriate assignment and task tracking. Throughout the operations life cycle, operational products will need to be developed and updated. Git will be used to track product development and to maintain a configured operational product repository. Updates to operational products will be tested and reviewed at a Configuration Control Board (CCB) meeting. New products or changes to existing products approved by the CCB will be merged into the operational product branch and pushed out to LMOC systems.

Products that are being created or updated often require testing against a spacecraft or observatory simulator. Both the L8 and L9 satellites have a designated Softbench and S/OS simulator. The Softbench simulator is a software only simulator which runs spacecraft flight software tasks and models flight software task response to provided commands. The S/OS is a higher fidelity simulator, which contains engineering model C&DH components, a partial solid state recorder, an OLI simulator, and a TIRS simulator. Data and log outputs from the simulators can be collected by Mercury and deposited on the CR for archive, review, and dissemination to external entities. Telemetry archives from simulator runs can be injected into STARS for analysis and plot generation.

4.2.5. Simultaneous real-time T&C with multiple spacecraft

The L8 and L9 orbits will be offset by 180 degrees, such that a single ground station would never be in view of more than one spacecraft at a time outside LEO&C. Though rare, it will be possible for both observatories to be in contact with different ground stations at the same time. The Galaxy T&C system allows for the LMOC ops engineers to receive telemetry from multiple spacecraft simultaneously by using the Galaxy context capability. The LMOC will use separate Galaxy contexts for each L8 and L9 observatory and simulator, along with a fleetview context showing both observatories simultaneously.

The Real-time Operations Consoles will be configured with access to only the L8 or L9 observatory contexts. Each will run two independent schedule executors that will automate spacecraft contacts.

Multiple mechanisms prevent an ops engineer from inadvertently transmitting a command to the wrong spacecraft:

- Real-time Operations Consoles are constrained based on the physical cabling to the appropriate mission encryptor.
- Command definitions will include the target spacecraft ID. If, somehow, a command is sent to the wrong observatory, it will be rejected.
- Pass procedures will verify the correct spacecraft ID is being received in telemetry and warn operators if the wrong ID is being received.

Each telemetry display will be color-coded and named to reinforce to the ops engineers the spacecraft with which they are in contact.

The fleetview context will be configured such that it will see telemetry from L8 and L9 together, but will not permit access to spacecraft commands. In fleetview, ops engineers can monitor all Out Of limit (OOL) violations, event messages, ground links, and telemetry from each observatory.

4.2.6. Instrument Calibration and Off-Nadir Slew Planning

A requested instrument calibration schedule is created by the Calibration/Validation team and reviewed against observatory constraints, operational limits, and mission planning capabilities. Iterative updates and review result in a stable calibration schedule, which is then added to the routine mission operations planning calendar. Calibrations are scheduled using the COST tool as Calibration Collection Requests (CalCRs). Ground Look Calibrations (GLCs), and Side Slither calibrations are exceptions and come in as Special Collection Requests (SCRs) through the User Portal. CalCRs do not require Data Acquisition Manager (DAM) approval, whereas SCRs do. If ad hoc, non-routine calibrations collect are required, the CalVal team raises them in the weekly planning meeting. CalCRs and SCRs are reported in the IDSR alongside LCRs.

Most calibrations are relatively straight forward and rely on onboard calibration sources. Onboard calibration rules are coded into CPAW for routine scheduling based on the mission operations planning calendar. Several calibrations are more complicated and require observatory slewing to acquire specific targets or light sources. These calibrations are still scheduled through the mission operations planning calendar, and for Lunar calibrations supplemental products are needed from FDS by CPAW to plan the event.

Flight Dynamics will supply mission planners (and CPAW) with an attitude maneuver plan and predictive attitude history file that matches the required lunar calibration attitude slew profile. The maneuver plan includes the desired slew profile that has been constructed to meet observatory pointing, thermal, and slew constraints. CPAW integrates the slews provided by FDS with slews calculated by CPAW to produce a complete attitude profile. This complete profile is then used to create a MAL containing the full set of calibration commands. CPAW also sends the complete attitude profile to FDS for use in the OBC validation process. Due to the additional slew complexity, Lunar calibrations require Flight Dynamics to create a separate

AWT table for direct spacecraft execution of the desired slew profile, which is sent to Galaxy and used to create spacecraft loads in parallel with the MAL.

4.2.7. Mission Data Management

Both L8 and L9 operate on a closed loop Mission Data Management cycle. Mission Data Management (MDM) is the process of scheduling, recording, downlinking, and clearing data from the recorder. MDM is a systems concept which relies on mission planning, onboard stored commands, real-time ground automation, station file reports, LFAT, and various LMOC products. L8 and L9 both utilize the MDM process, however each satellite's MDM cycle and MDM products are handled independently within the LMOC.

4.2.7.1. Overview

- 1.) Science scenes are scheduled by the USGS DAM. The acquisition plan and approved special scene requests are transferred from DPAS to the LMOC
- 2.) The acquisition plan and special scene requests from DPAS are used by Mission Planning to schedule scenes within the daily observatory command load products. In addition to the daily command load products (ATS/IWT), the SIMT file is generated which contains the scenes scheduled to be imaged during execution of the load and their mapping to root file ID within the SSR. The daily loads are verified by the operations team and uplinked to the observatory.
- 3.) The observatory executes the daily command load and records scenes from both the OLI and TIRS instruments as per the stored commands. Recorded instrument data are saved as root files and sub files on the recorder. Spacecraft stored state of health (SSOH) data are recorded and saved to designated root files on the recorder.
- 4.) During RT contacts, recorder files are downlinked via CFDP protocol to the ground station based on priority. A recorder directory listing is dumped at the beginning and end of the pass to show the current state of the recorder.
- 5.) Software at the ground station determines if the files are complete. If complete, the file is given an ACK (Acknowledged). If incomplete, no record is created.
- 6.) The station sends the GPR (Ground Post-pass Report) to the LMOC that states the files have been received from the contact.
- 7.) The GPR (status of files received during the pass), SIMT (expected files to be received during the pass), and recorder directory listing (file status on the spacecraft at the end of the pass) are compared within LFAT. Files which are accounted for and complete are marked for unprotect on the recorder and deletion. Files which are missing or incomplete are marked for retransmit.
- 8.) Steps 3-7 are repeated on subsequent contacts. In addition:
 - a. On Landsat 8: Ground automation builds ROSs (Relative Operational Sequences) which contain files marked for unprotect/deletion for uplink on the next pass and execution in the back-orbit. ROSs are generated from the unprotect/delete list from LFAT.

On Landsat 9: Ground automation builds unprotect/delete commands that can be uploaded to the spacecraft and executed in real time. The unprotect/delete commands are generated by LFAT.

b. On Landsat 8: Ground automation identifies files transmitted as real-time images during a pass and builds and uplinks a ROS to mark these files as played back to be executed in the back-orbit.

On Landsat 9: Ground automation identifies files transmitted as real-time images during a pass and builds and uplinks commands to mark these files as played back in real-time.

c. Ground automation retransmits missing or incomplete files via ground command during RT supports

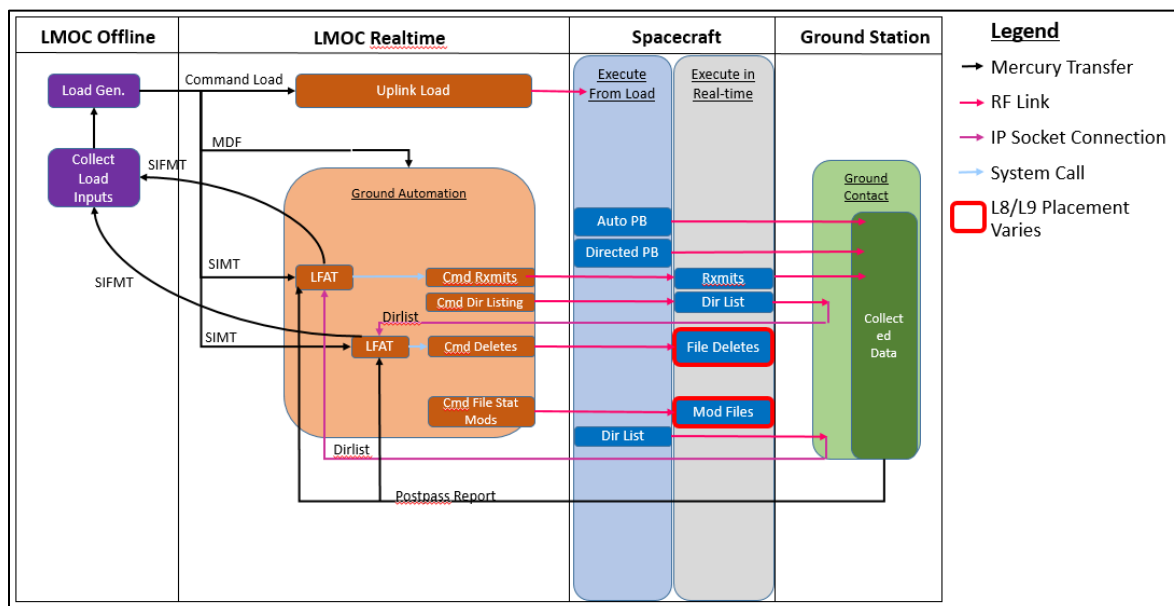


Figure 4-9 MDM Overview

4.2.7.2. Data Recording on an Observatory

The most significant MDM-related action on the observatory is the recording of new Mission Data. The spacecraft maintains a file system for storage of mission data and SSOH. Files are named with a 3 digit root file ID and a 3 digit subfile ID. One interval of normal science data (looking at the Earth) will consist of a recording of OLI data and a recording of TIRS data. Each recording is assigned a single root file ID. MPS assigns these root file IDs as part of the load building process. The first 50 root files are reserved for SSOH data, where the remaining root files can be allocated to either TIRS or OLI data (note: on Landsat 8 the last root file in this range has been reserved for the TIRS throwaway file).

When the observatory records data from either instrument, it always starts with subfile 000. When this subfile reaches 1GB in size, it starts recording to subfile 001. This process continues until the command to stop recording is encountered. Since OLI generates so much more data per second compared to TIRS, there are usually many more OLI subfiles than TIRS subfiles for long intervals. The longest TIRS interval capable of being recorded by the observatory would only have 3 subfiles.

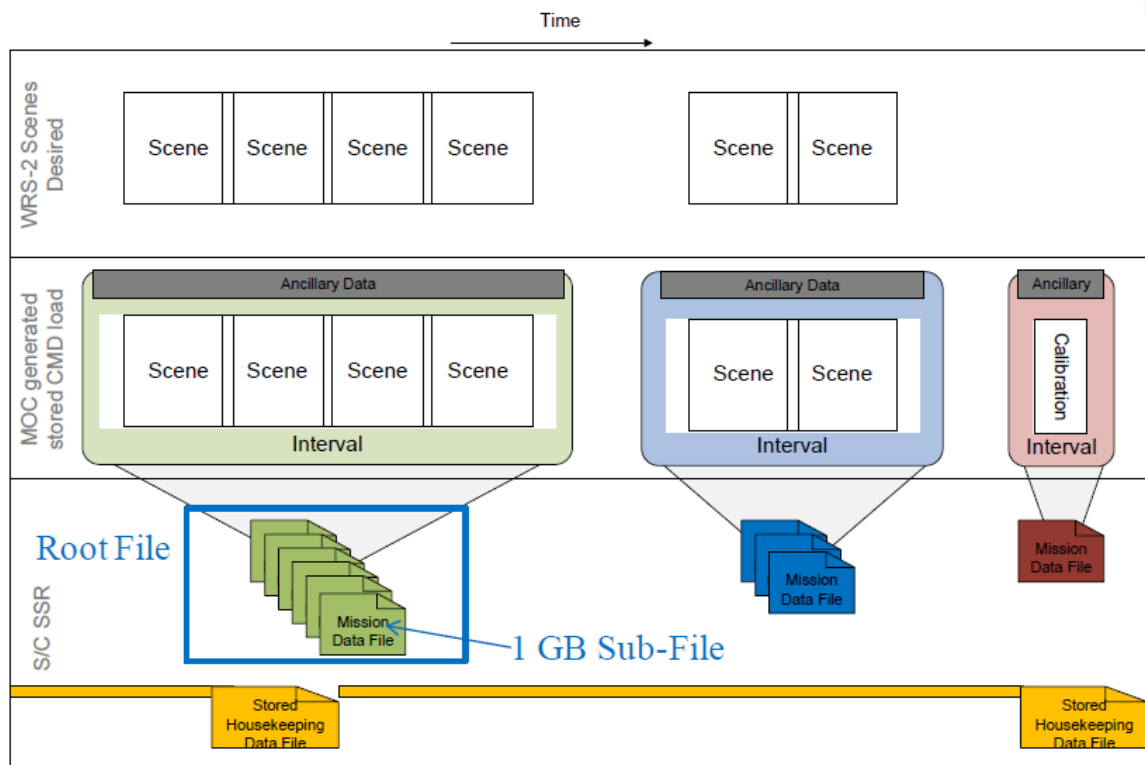


Figure 4-10 Activity Planning

Landsat 8 TIRS data has a known issue where some CFDP packets going to the SSR are placed in the wrong subfile. The mixing of a portion of data between subfiles means that when individual TIRS subfiles are downlinked, a portion of the subfile is essentially missing (since it is in the next or previous subfile). This creates a persistent downlink failure when individual TIRS subfiles are downlinked. When TIRS files are downlinked by rootfile this issue does not occur since the CFDP protocol is checking that data is present for the whole root file. Landsat 8 TIRS playback and retransmits are only to occur via root file.

Data is recorded to the SSR as an interval with the **Protected** attribute set to true. (On Landsat 8 overwriting files can be done, but was not recommended by the vendor due to large variances in command execution times). To minimize the number of commands required for recorder management, files ready to be deleted are marked as unprotected and then deleted by a 'delete all unprotected files' command (completed via ROSs in the back-orbit on Landsat 8). Deleting a subfile frees up the recorder space and that subfile number for reuse. The root file ID is not freed up until ALL subfiles using that root file ID are deleted.

4.2.7.3. In Pass MDM

Mission Data Management (MDM) performs data retrieval and recorder maintenance. MDM is coordinated between onboard IWT commands (uplinked daily), ground commands, LFAT, and ground automation/procs. (Note: on Landsat 9 there is no explicit IWT command buffer,

however a set of ATS buffers will be referred to as the IWT buffer set for operational continuity between L8 and L9).

Prior to each GNE contact a command in the stored command load builds the auto-playback list. FSW identifies files not marked as played back, and sorts them by priority. Within each priority group files are sorted oldest to newest. An observatory stored command scheduled to occur after X-band AOS begins autonomous playbacks. When autonomous playbacks are issued, FSW internally generates a list of up to 14 files based on the first 14 entries of the auto-playback list and begins file playback. If there are more than 14 un-played back files in the auto-playback list, FSW will create a new 14 file playback list after the playbacks of the first 14 files have completed.

While an autonomous playback is in progress, a stored command is executed for a directed playback of housekeeping data (SSOH file). SSOH files contain the engineering data used to generate sequential prints, engineering plots, and are also used to alert the FOT of back-orbit limit violations. Directed Playbacks take priority over Autonomous Playbacks and the autonomous playback already in progress will be suspended until the directed playback is complete.

After X-band downlink has been verified by the ground system, ground automation issues a directed playback command for the retransmit of any missing files from prior passes. The reason retransmits are issued via ground command is so they can be uplinked to the spacecraft in an expedient manner. The retransmit list used for the retransmit command is generated as an LFAT output.

Once any necessary retransmit commands are uplinked, ground automation will command a dump of the FSW directory listing. On L8, this is done at the beginning of the pass so that the unprotect and delete commands executed via ROS in the prior back-orbit will be reflected in the listing. The directory listing is then compared to the GPR report present from the last support in LFAT and all accounted for files are marked for unprotection. On L8 the unprotect and delete commands are then packaged in one or more ROSs and uplinked to the observatory. The ROSs will then execute in the back-orbit when there is no scheduled imaging. (Simultaneous recorder file I/O commanding has caused SSR commanding issues). The process is simplified on L9; unprotect/delete commands do not need to be packaged in a ROS and can be uplinked and executed in real-time.

Ground automation tracks real-time images being downlinked during the pass. Near the end of the support, ground automation will make status modification commands for these files that will change the real-time downlinked file status from ‘unplayed back’ to ‘played back’. On L8 the status modification commands are wrapped up in a ROS and uplinked to the observatory for execution in the back-orbit. L9 does not require packaging these commands within ROSs and can be executed in real-time. The status modification of the real-time downlinked files prevent FSW from including the real time images downlinked during the pass in the next support’s autonomous playback (which would double downlink the files acquired during real-time imaging). The station GPR will show if the file was successfully captured. If ACKed, LFAT will mark the file for deletion. If not received, LFAT will mark the file for retransmit on the next support.

Prior to LOS, stored commands stop autonomous playback and issue a directory listing. The directory listing will be used to compare what files (and file states) are on the recorder for comparison to the GPR report and the SIMT. Output from this comparison will be used to generate the next support's retransmit list.

4.2.7.4. Data Accounting

Once a file is played back or retransmitted to the ground from the observatory via the CCSDS CFDP protocol, the ground station generates a Ground Post-pass Report (GPR). The GPR contains a list of files from the observatory which were successfully received by the ground during the pass.

When the GPR is delivered to the LMOC, the SIMT (which has the anticipated list of files that were to be recorded), GPR (contains which files were received on the ground), and the directory listing (the state of the recorder) are compared within LFAT. LFAT then creates the following outputs:

- Scene Interval File Mapping Table (SIFMT): captures the current state of the recorder for use by MPS. File IDs which are open in the SIFMT can then assigned to scenes in the next IWT. The SIFMT is also sent out the DCRS that uses the file to link scene/intervals to the file numbers for interval assembly and processing.

- Retransmit Commands: LFAT generates a list of files that were missing in the GPR. Ground automation references this list for directed playback during the next RT contacts.

- Unprotect Commands: Ground automation uses this list to create the unprotect and delete list for files confirmed as being received on the ground. L8 uses this list to generate the unprotect and delete ROS, whereas L9 directly uses this list to unprotect and delete files by real-time command

- Delete Commands: Due to the current ops con of unprotecting files individually and issuing a delete all unprotect, this list is not used but is maintained by LFAT.

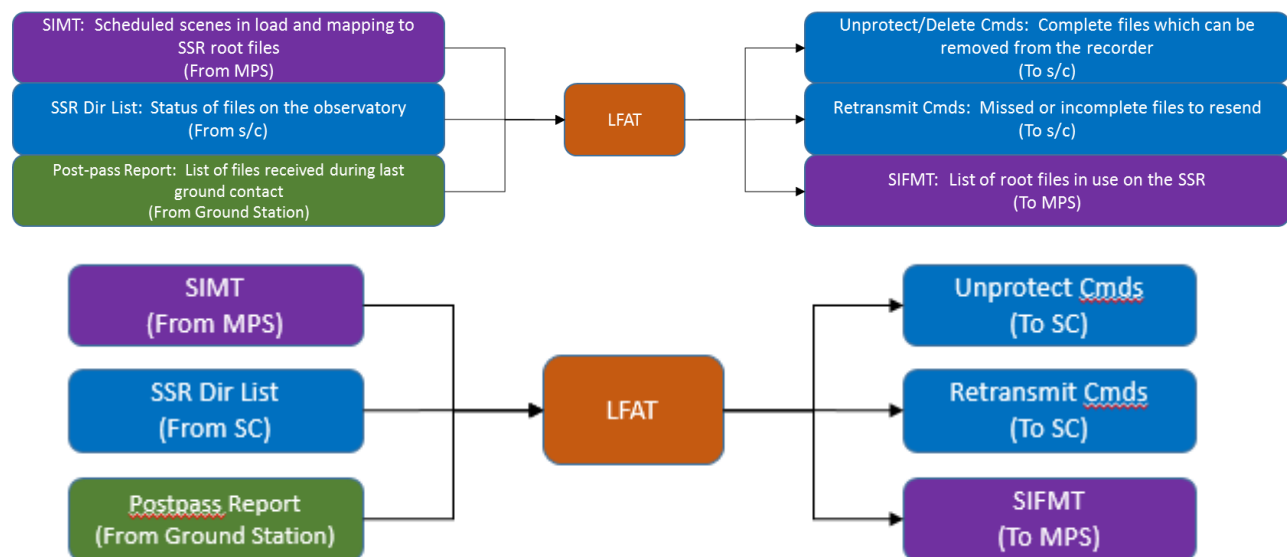


Figure 4-11 Data Accounting

4.2.7.5. Pass Automation

Several ground automation procs are used to prepare for and execute a real-time pass (and MDM). Kicking off the entire pass automation sequence is the MDF. The MDF is referenced by Schedule Executor on the Real-time Operations Console (ROC) to identify upcoming contact AOS and LOS times. Prior to AOS the pass proc is called that begins configuration of the ground system. If the L8 pass proc is run with the “FULL” option, the pass proc also calls code which generates MDM ROSs for unprotecting and deleting files after the next pass.

Once the pass begins and X-band playbacks are active, the ground automation knows the spacecraft has been acquired by the ground station and proceeds to issue a direct playback command for retransmits. Ground Commands are coordinated with stored commands on the observatory to create the desired autonomous playback, directed playback, retransmit, directory listing, unprotect/delete, and file playback status mod timing. After the retransmit proc completes, ground automation will issue unprotect/delete and file playback status/double downlink mitigation mod commands or ROSs.

At the end of the pass the MDF calls the post-pass proc, which initiates several pass cleanup activities. Pass automation timing may differ slightly between L8 and L9, but the overall command sequence and cadence will be the same. L9 recorder command execution times are being tested.

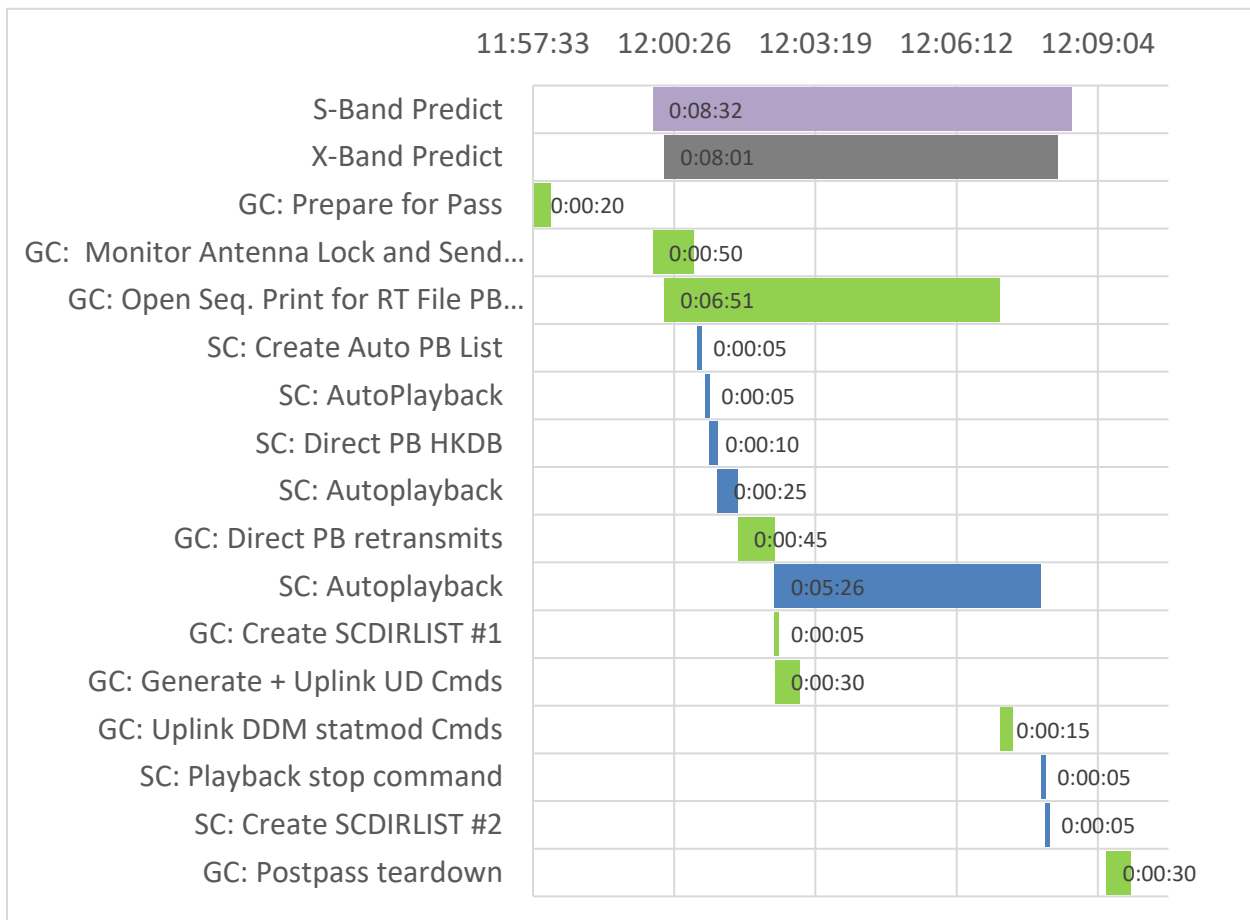


Figure 4-12 Pass Automation Timeline

4.2.8. Autonomous operations and notifications

LMOC automation will use computers to control and perform daily/weekly routine activities that include mission planning, real-time operations, offline processing, data management, and remote notification functions. Automation must meet a 72 hour lights out (no FOT intervention) requirement (LMOCD-2423). This includes both the spacecraft and the ground system. Spacecraft automation will use the stored command loads, auto-playback MDM feature, and the ability of Galaxy to start automated STOL procs for observatory commanding. Ground automation will use the ROC ability to start processes using a directive file. All non-nominal and critical activities will be staffed and critical products will be tested and reviewed by operations before use.

There are no LMOC requirements to perform automated failovers of ground equipment. The operations team will have to support off-hours ground system anomalies to get the systems failed over to the backup units. The spacecraft, on the other hand, will use fault monitoring to safe and in some cases correct problems.

LMOC subsystem feature varying levels of automation which combine into a decentralized LMOC automation system. Key LMOC automation aspects (Figure 4-13) are very similar to the legacy L8 MOC and include:

- Product/file transfer
- Data Archive
- Pass operations
- Data accounting
- Observatory alerts
- Batch plot generation
- LMOC/bLMOC subsystem alerts

While Mission Planning and Flight Dynamics daily processes will not be completely automated, the LMOC will have significantly more automation in these areas to expedite daily analysis and product generation. Much of the LMOC automation is described in sections 4.2.3 and 4.2.5.

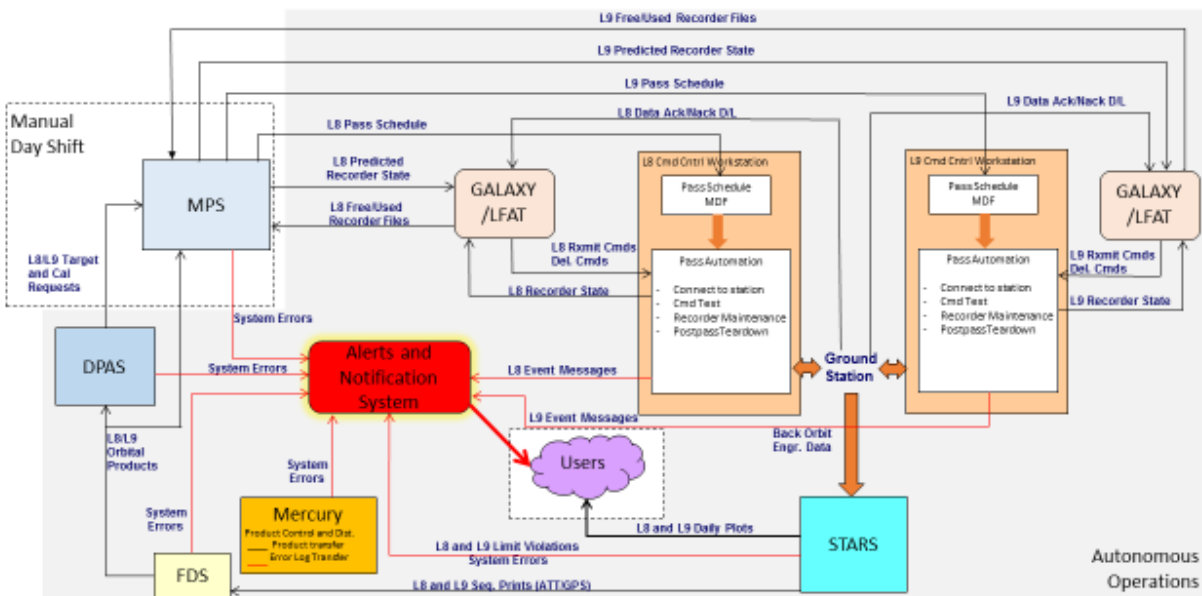


Figure 4-13 LMO System Automation

The Mercury subsystem is the heart of automation and controls data distribution within the LMO. The Mercury system uses a configurable product flow model and remote agents to:

- Recognize the arrival or creation of file-based data products.
- Move /copy products from source to destination, and can initiate processing at the destination.
- Identify failures in product flow and production.
- Maintain and provide controlled user access to product repository.

These Mercury functions are used to automatically move files between LMO subsystems when appropriate conditions are met. Mercury is also used with the Alert and Notification System to issue data product alerts if products are not received by user-defined deadlines. Mercury also transfers subsystem and observatory limit violations and error logs to the Alert and Notification Subsystem for team notification. Important products can be flagged in the Mercury model for copy to an archive or a backup location.

A substantial amount of automation is used in pass operations and data accounting as described in section 4.2.5. The automation cycle between CPAW, LFAT, observatory, Stars, and the ROC/Procs allow for automated ground contact configuration, new data downlink, data retransmits, file deletes, file status modifications, ground contact cleanup, data accounting, observatory transmitter management, and observatory data collection.

After an L8 or L9 ground contact observatory stored state of health (SSOH) data from the last orbit is transferred to the LMO. When new SSOH data is received, STARS automation is used

to verify the SSOH integrity and ingest the data into the STARS data archive. STARS will capture any observatory limit violations observed during data ingest and create a file for transfer to the Alert and Notification Subsystem and operations team notification. STARS also utilizes automation to generate L8 and L9 batch plots for spacecraft engineer analysis, and sequential prints with attitude and position data for daily FDS use.

Observatory limit violation, system logs, and system error files are transferred to the Alert and Notification System (ANS) by Mercury. The ANS uses local automation to parse received logs, limit violations, and events for predefined error conditions. If an error condition which has been deemed to warrant user alert has been met, the ANS then notifies the applicable operations or user group via email and/or SMS. Alerts can also be generated for loss of subsystem heartbeat (indicating a subsystem issue), or problem with automation file delivery within a predefined amount of time.

4.2.9. Orbit maintenance maneuvers

Orbit maneuvers will be performed on a regular basis to adjust the orbit altitude (DMU) or inclination (INC) in order to maintain the observatory's ground track and MLT, respectively. The orbit ground error, measured at the descending node crossing, must stay within 5 kilometers of the WRS-2. The mean local time of the L8 orbit is subject to a constellation pole crossing constraint of a minimum MLT = 10:10am.

4.2.9.1. Drag Make-Up Maneuvers

Drag due to atmospheric conditions will dictate DMU maneuver frequency to maintain the ground track within limits. To account for orbit prediction errors, variations in the predicted solar flux data, and orbit maneuver errors/uncertainties, the operational ground track error is maintained from 0 to +3 km in order to ensure that the requirement is met. This also reserves 0 to -5 km as a buffer for RMMs. DMU maneuvers will be predicted automatically by the LMOC FDS component during the daily predicted orbit and product generation functions and modeled in the planning products. Additionally, the location in the orbit for the burn will be optimally chosen to maintain a frozen orbit (the proper eccentricity and argument of perigee). DMU times are announced ~3 weeks in advance in order to schedule the appropriate communications coverage. The maneuver magnitude will be finalized commensurate with the load build for the day of the burn, approximately 2 to 3 days prior.

The maneuvers are sized conservatively to maintain the ground track within the required limits, accounting for variability in the system performance and predicted solar flux (drag) predictions. The burns have these general characteristics: 1-second duration, 0.04 m/s delta-v, and a 70-90-meter altitude increase.

4.2.9.2. Inclination Control Maneuvers and Constellation Coordination

Inclination maneuvers must be periodically performed to maintain Mean Local Time (MLT) to meet data collection specifications, 10:00am +/- 15 minutes. Inclination maneuvers are coordinated with NASA Earth Science Mission Operations (ESMO) Earth Science Constellation (ESC) to ensure there will be no close approaches between Landsat missions and other 705 km satellites operating within close proximity. As such, L8 is subject to an ESC pole crossing constraint of a minimum MLT = 10:10am. Inclination maneuvers are performed on a yearly basis to maintain the orbit MLT within an operational range of 10:11 – 10:12:45am. A typical inclination maneuver requires 5-6 m/s delta-v and adjusts inclination by approximately 0.03 to 0.04 degrees. The maneuver duration for these burns is around 60-90 seconds.

4.2.10. Conjunction Assessment and Risk Mitigation Maneuvers

A risk mitigation maneuver (RMM) is executed in response to conjunction assessment (CA) screening results where a conjunction with another spacecraft or piece of debris presents a high risk for collision. The magnitude of such a maneuver can vary with each conjunction's circumstance and environmental conditions and should avoid violating the spacecraft control box if at all possible.

When a potential high risk conjunction is identified the CA team works with the Project and operations to explore possible options. Several iterations and burn sizes may be planned and screened by CA service provider to determine if the outcomes will reduce or eliminate the risk – or to unintentionally introduce new collision risk. Because the results are more accurate as the conjunction time approaches, an option to maneuver is typically finalized very close to the burn time (sometimes within 1 day).

Both ephemeris, including covariance, and maneuver data will be provided to the CA service provider. Routine CA analysis will be performed for the L8 and L9 missions utilizing data provided by the United States Strategic Command (USSTRATCOM) Joint Space Operations Center (JSpOC). The process consists of:

1. LMOC provides predictive L8 and L9 ephemeris with covariance (burn and no-burn), based on orbit determination solutions from the GPS data, or FDF during LEO&C.
2. The L8 and L9 predictive ephemerides with covariance are delivered to the CA service provider via sFTP.
3. The CA service provider, in turn, delivers the ephemeris to JSpOC for screening analysis.
4. JSpOC identifies close conjunctions between L8/L9 and other orbiting objects in the USSTRATCOM Space Objects Catalog, based on comparisons of the ephemerides of objects contained in the catalog with the satellite ephemeris and with the JSpOC-produced L8/L9 ephemeris.
5. The predicted time of closest approach and miss distance, along with other quantitative characteristics for the close encounter, are passed back to the CA service provider for further, in-depth analysis to quantify the risks.

-
6. For each predicted event, the CA service provider determines the nature of the threat (i.e. the risk of collision) posed to the L8 or L9 spacecraft, including the probability of collision and other relevant geometrical factors intrinsic to the conjunction.
 7. If the conjunction is deemed to be a “high interest event”, the CA service provider consults with the LMOC operations to discuss recommendations and to determine and plan any necessary mitigating action.

L8 CA services are provided by SpaceNav. L9 will utilize the GSFC Robotic Systems Protection Program (RSPP) Conjunction Assessment Risk Analysis (CARA) team for CA services for LEO&C (cutoff would be the Mission Transition Review (MTR) or Post Launch Assessment Review (PLAR), which should occur on the same day or successive days at around L+90). Afterwards, SpaceNav will provide L9 CA services in addition to that of L8.

The nominal prediction frequency with full screenings is 3x daily. High risk conjunctions or special request ephemerides are also assessed out of cycle on evenings, weekends, and holidays as needed. Predictions are made 7 days into the future.

4.2.11. Simulator operations

The LMOC will have designated simulators for both L8 and L9. Each satellite will have one or more Softbench simulators and a Spacecraft/Observatory Simulator (S/OS) provided by the spacecraft vendor. Each satellite will also have a Software Development and Verification Facility (SDVF) high fidelity simulator which will be located at the spacecraft vendor facility and maintained by the spacecraft vendor. Since operations will not be tasked with the direct operations or maintenance of the SDVF, it is out of scope of the OpsCon.

Softbench Simulator Overview:

- Used for FSW development, I&T procedure/script development, and Operations procedure development and test
- Software-based emulation of Spacecraft
- Runs actual spacecraft FSW
- Softbench is a software simulation of SC built within commercial SW (SIMICS, Wind River):
 - SW models of SC HW including RAD750 CPU
 - Actual Flight Software (FSW) runs on RAD750 CPU model
 - SW models of ACS, EPS and TCS dynamics (ACS dynamic aka plant model)
- Softbench is part of FSW development/testing prior to FSW testing on SDVF

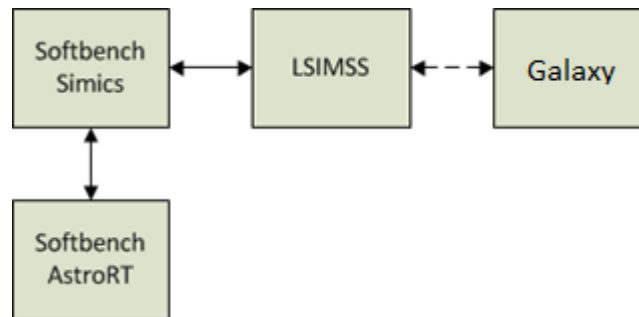


Figure 4-14 Softbench Interface Diagram

S/OS Simulator Overview:

- Used at LMOC for ground system testing, FOT training, procedure verification, and anomaly resolution with interfaces to Instrument Simulators/ GALAXY/LSIMSS
- SOS is comprised of the Northrup Grumman Innovation Systems (NGIS) components plus the GFE components (OLI Simulator, TIRS Simulator, GALAXY/LSIMSS).
- Runs actual spacecraft FSW
- S/OS can be commanded using AstroRT or GALAXY (only one T&C system can control)
- S/OS can have simultaneous multiple telemetry monitoring (AstroRT and/or GALAXY)
- S/OS provides high fidelity simulation with some flight-like hardware (EM-PIE, EM-SSR)
- Image data is passed from Instrument Simulators via HSSDB into the PIE and passed on for WDC evaluation.

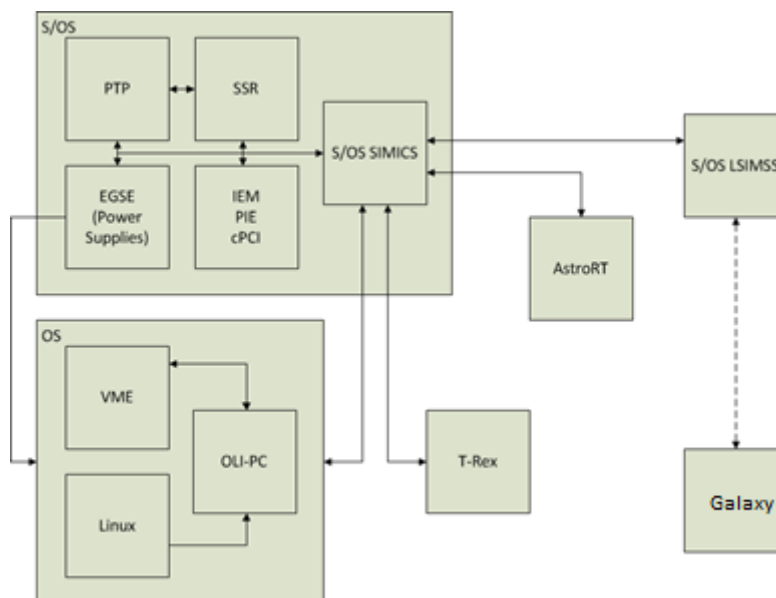


Figure 4-15 S/OS Interface Diagram

Support Software:

SIMICS – Simulation software used to run observatory subsystem (FSW, EPS, TCS, etc.) models

- Includes models for FSW, EPS, TCS, ACS/orbital dynamics, and thrusters
- Does not include science instruments and most comm equipment

ASTRO RT – vendor supplied telemetry and command software used to interface directly with SIMICS. Used to issue simulator configuration commands and to inject faults.

- Sends commands, receives and displays telemetry, produces archives
- Also sends metacommands to simulator, like checkpoint writing
- Interfaces directly with Simics, no middleman

LSIMSS (Linux Scalable Integrated Multimission Simulation System) – Provides telemetry and command routing and formatting between Simics and LMOC GALAXY/ROC.

- Provides telemetry and command routing between Simics and GALAXY
- There is one per simulator

The Landsat simulators will be integrated within the LMOC and will interface directly with LMOC subsystems. The simulators will be used by the FOT to simulate interactions between the LMOC and the flight L8 or L9 observatory. Simulators will receive and executes commands, and generates simulated telemetry for analysis by FOT personnel. The higher fidelity S/OS simulators are to mimic capabilities of the observatory and its subsystems - including the imaging sensor(s).

The simulators will be used, at a minimum, for the following:

- FOT training
- Procedure validation
- Maneuver rehearsals and validation
- Flight software updates testing and validation
- Anomaly investigation and resolution

Simulators will be subject to the same CM protocol as the spacecraft to manage FSW and configuration file versions. Upon simulator startup configuration scripts will be run to load the current operational products to the simulator and to step through the process of configuring the simulator to the desired operational state. Simulators can be set to use current wall clock time or past/present time as needed to match the desired simulation scenario. Long duration sims will likely require simulator engineer care and feeding to keep subsystem models and timing true to life.

Other than to resolve functional issues after delivery, Softbench core SW, AstroRT and EGSE SW will not be updated after delivery. Functional issues are:

-
- New release of C&T Database (if functionality affected)
 - Minor changes (e.g. conversion factors) may only affect AstroRT
 - Major changes would require replacement of FSW
 - New release of FSW (if functionality affected)
 - May require new functionality within Softbench or EGSE in addition to replacement of FSW

During prelaunch testing one or more operations team members will be tasked with simulator checkout, configuration, and maintenance in support of MRT and GRT testing. After launch the operations team will have relatively unrestricted access softbenches for operations development and testing. Since the softbenches are software models, the softbenches are very robust and do not require tight oversight. The S/OS simulators however are more sensitive to misconfiguration and should always be configured and maintained by trained operations staff. The S/OS components are ESD sensitive and require ESD certified personnel for handling.

Simulator training documentation, diagrams, and component lists are to be delivered to the operations team from the spacecraft and instrument teams upon simulator delivery. Since the simulators are complicated, onsite training or vendor support may be required to properly equip the operations team for sustained simulator maintenance and operation.

4.2.12. Mission Analytical Modelling

Interrelation of recorder operation, recorder resources, data collection, calibration schedules, and station coverage create difficult analysis and optimization problems for L8 and L9. The Analytical Model is designed to provide strategic analysis of mission baseline changes in these areas. Analytical Model simulations can be used to determine recorder margins for different station configurations or data collection schemes, calculate file latency to DCRS, and to model system response to typical MDM, ground station, or DCRS problems.

The Analytical Model provides simulates the Landsat mission in the following areas:

- Landsat 8 data collection
- Landsat 8 recorder resources
- Landsat 8 mission data management
- Landsat 8 ground network
- Landsat 8 data playbacks
- Landsat 8 data collection
- Landsat 8 recorder resources
- Landsat 8 mission data management
- Landsat 8 ground network

-
- Landsat 8 data playbacks
 - Mission Data transfer from stations to DCRS
 - Mission Data ingest and processing through DCRS

The Analytical Model uses the GOTS JASON Landsat model software developed by the USGS for use in Landsat 9 ground system requirement verification.

The Analytical model uses a generic ‘mission model’ which can be configured with parameters to simulate aspects (noted above) of either the L8 or L9 mission. If the user desires to include both L8 and L9 in a simulation, the mission model is run with the user supplied L8 parameters, then run again with the user supplied L9 parameters. Results from the L8 and L9 mission model runs can be used by the FileTX module to calculate transmission time of data to the USGS DCRS. The results of the FileTX module can then be used as inputs to the DCRS module to calculate estimated processing times through DPAS. Inputs and outputs to all modules are stored within a MySQL database for analysis. Each module is coded in C++ and can take user supplied inputs to configure the module for the desired simulation. The Analytical Model user can also pass inputs into a simulation to model outages or random failures of key mission systems (ground station, LFAT, WAN, DCRS processing strings, etc.).

The Analytical Model can be used within the LMOC from the Mission Planning System since it will share the MPS VM. Since the operations team and USGS team members may want to use the Analytical Model, the latest operational version of the Analytical Model code will be hosted on the USGS Bit Bucket. Users (FOT or USGS Ground System Engineers) with the correct privileges can download the latest operational code branch from Bit Bucket for installation and use on their local USGS workstation or laptop. Any Analytical Model updates developed on systems external to the LMOC can be submitted to the USGS Bit Bucket for operational testing and possible incorporation into the operational code baseline.

4.2.13. Training and certification

The LMOC Operations Team will be trained and certified to operate the mission. Training will include:

- Functionality of the observatory (spacecraft and imaging sensor(s))
- Functionality of LMOC software and hardware
- Operations procedures including LEO&C, nominal operations, contingency/recovery procedures

Training will take various forms, including exercises to simulate typical day-in-the-life on-orbit Observatory operations and launch rehearsals. To the extent possible/practicable, the operations team will be trained using the system(s) and interfaces that will be used in actual operations. Where this is not possible, equipment that emulates the interfaces and simulates performance

with high fidelity will be used. Each operations position will have an associated training and certification plan monitored and administered by the FOT manager and system leads.

Certain LMOC subsystems have a repeatable set of daily production steps or processes:

- Flight Dynamics
- Mission Planning
- Ops Engineer

These positions will have step by step training procedures and certification processes to ensure new trainees are familiar with, and prepared for, these positions.

Other subsystems within the LMOC have fewer standardized/daily processes and rely on a body of experience for troubleshooting:

- Mercury
- STARS
- Galaxy
- Alert System

Such subsystems will be managed by Ground System Engineers. Ground System Engineer training will still utilize training documentation and procedures, but will incorporate substantially more on-the-job training.

Observatory engineers are tasked with tracking observatory health and safety and resolving onboard anomalies. Observatory engineer training will require thorough review of subsystem SOPs and COPs. Simulations of common (or feared) observatory anomalies will be utilized to provide observatory engineers with additional training and experience.

Infrequent but important operations activities (TDRSS proficiency, Blind Acq, Neg Acq, NEN contact, LMOC subsystem failover, failure to bLMOC) will be exercised periodically to maintain operations team proficiency. Prior to executing a proficiency task, the team will coordinate and schedule the activity so that nominal operations are not disrupted.

The Operations Staff Training and Certification Plan (OR-3) is CM controlled and will be updated as needed throughout the mission life cycle.

4.2.14. Anomaly response

The operations team will be alerted of anomalies by either real-time system use during shift operations, or by automated alerts from the Alert and Notification System. If an anomaly is observed during work hours, the observing engineer is to report the anomaly to the applicable system lead (observatory, real-time, ground system, or system administration) for further

investigation and possible escalation (Figure 4-16). After being notified of an anomaly the system lead will determine if the anomaly is minor or major.

Minor anomalies are localized to a subsystem and do not impact observatory health and safety or other system functions. Typical minor failures may include subsystem failover, delayed product receipt, or a missed ground contact. Minor anomalies are resolved as quickly as possible during work hours and reported in the following daily standup and the applicable management reporting cycle. Known or reoccurring minor anomalies identified by operations and agreed to by the USGS FSM can be immediately worked/resolved without escalating to a system lead.

Major anomalies, however, are those that can jeopardize observatory health and safety and mission data and require immediate action. Major anomalies are reported by the system lead to the Flight Operations Team Manager who then notifies the USGS Flight Systems Manager. The Flight Operations Team Manager and Flight Systems Manager will alert external entities that may be impacted by the anomaly and to organize external support as needed. During a major anomaly the operations team will focus the immediate response on taking steps to ensure observatory or data safety and then collecting information to understand the anomaly scope and cause.

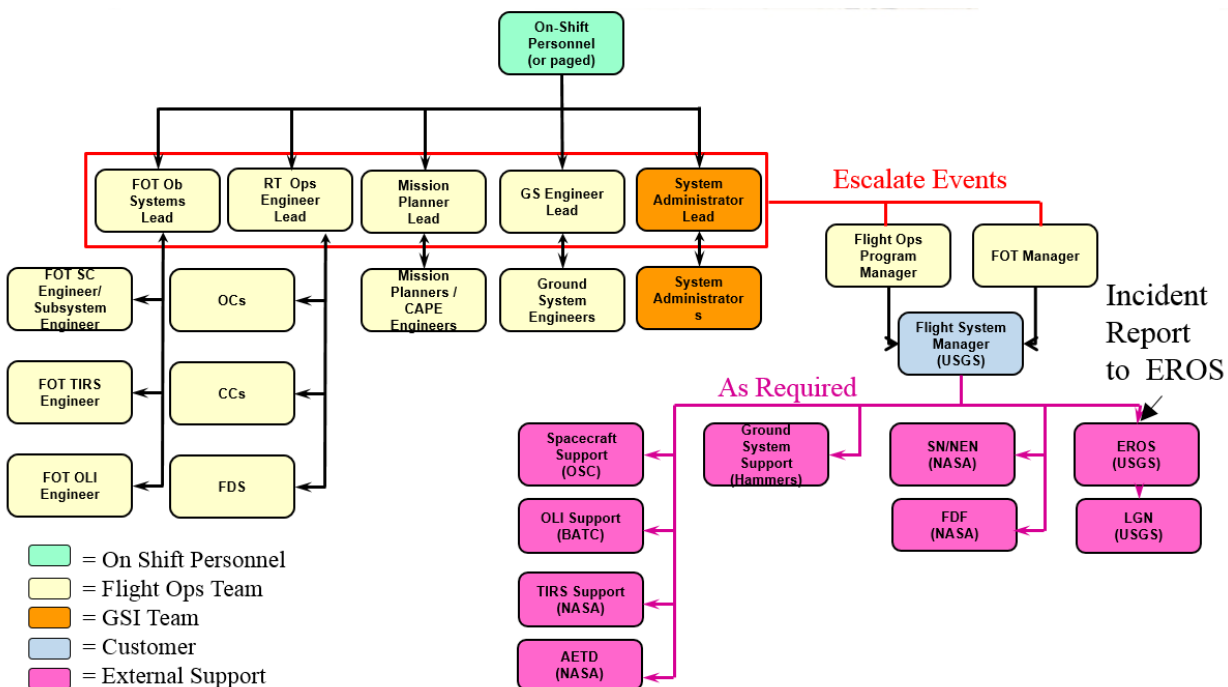


Figure 4-16 Anomaly Response Flow

Major anomalies will be handled by an Anomaly Resolution Team (ART) assembled by the USGS. The ART is tasked with investigating the anomaly, identifying root cause, and providing mitigation or recovery actions. The operations team will directly support the ART and will be responsible for providing engineering data for analysis and operations support.

The Alert and Notification System will generate notifications for anomalies experienced outside of work hours. The responding operations team member is to provide initial anomaly investigation and to escalate the anomaly response as needed. The operations team will provide a response to a major anomaly 24/7.

4.2.15. IT security patching & scanning

IT security patching and scanning is a periodic activity repeated during the LMOC lifecycle to help prevent malicious computer system exploitation. Patching is used to update LMOC system OS and application flaws to prevent exploitation. Patching can also provide software stability and performance updates. Scanning is conducted to identify system or software configurations which have known vulnerabilities. Patching and scanning policy fall under both the USGS and NASA guidelines and where policy differs the more stringent of the two shall be applied.

The patching and scanning cycle is based on USGS/NASA guidelines and the scheduling of LMOC systems for patching and scanning is handled by the operations lead. Prior to every patching and scanning cycle, the operations lead will create a bLMOC and LMOC system release schedule according to anticipated operations activities and system availability. If possible bLMOC systems are patched and scanned first to allow for operations testing prior to updating primary operations systems.

Prior to starting a patching and scanning cycle the system administration team identifies applicable Windows patches on the Windows Server Update Service (WSUS) and the Linux Redhat Satellite patch server. Identified patches are downloaded and staged by host for each bLMOC and LMOC system.

The day of system scanning and patching the operations team releases the system after system primary operations is completed for the day. If a system has no primary operations scheduled, it can be released for patching and scanning at start of business. Upon release to the administration team system patches will be applied. After the system baseline has been brought to the desired level, system scanning can occur to verify the anticipated system configuration and compliance with IT security (ITSEC) protocol. If issues are noted by the scan, the system admin will update the system configuration as needed and rescan. The system administrator will perform a reboot and verify operating system/box level functionality before returning the system to the operations team. Key updates to the OS environment or configuration will be captured and updated in the OS gold image for inclusion in the CM repository.

Systems handed back to the operations team will be reconfigured for operations (restart applications, automation, etc.) by the operations user and go through a system checkout. Any issues noted during checkout will be documented and reported back to the system administration team. Systems that pass checkout are then cleared for routine operations.

Patching and scanning can vary considerably from system to system, however under routine conditions a few hours are required and the system can be returned to operations the same business day. Due to minimal system outage, operations typically do not fail over to a redundant system during primary system scanning and patching.

4.2.16. Infrastructure sustainment

Routine care, feeding, and maintenance is required by LMOC system administrators to ensure smooth operation of the LMOC infrastructure. While periodic maintenance is rather minimal, a key role of system administrators is to monitor the LMOC hardware and software performance to spot and mitigate potential issues. Typical periodic and preventative maintenance work includes:

- Monitoring system resources
- Maintaining and updating firewall rules as needed
- Managing software licenses and extending software licenses as needed
- Managing hardware support agreements and extending agreements as needed
- Testing and installing pertinent OS or support application updates
- Monitoring system and network activity logs for suspicious activity

The ANS system can be configured to monitor for infrastructure or subsystem issues and to alert system administrator and ops team members if an issue or misconfiguration is identified. Identified issues are to be elevated as required as per the anomaly response plan.

The LMOC is engineered with a nominal 5-7 year hardware refresh cycle.

4.2.17. Configuration Management including deployment of CM-approved ops products

The Configuration Management of controlled items will be performed as per the process dictated in 02-PM-09, LMOC Configuration Management Plan (CDMP). This CDMP establishes the LMOC Configuration Control Board (LMOC O-CCB) and the process it governs, as depicted in the flowchart in Figure 4-17. Prior to launch, the O-CCB is jointly chaired by the NASA Mission Ops Readiness Manager (MORM) and the USGS Flight Systems Manager (FSM). Post-launch the O-CCB is chaired solely by the USGS FSM. Other participants of the LMOC CCB include the FOT, the LMOC administration team, LMOC vendors, and other experts, as needed.

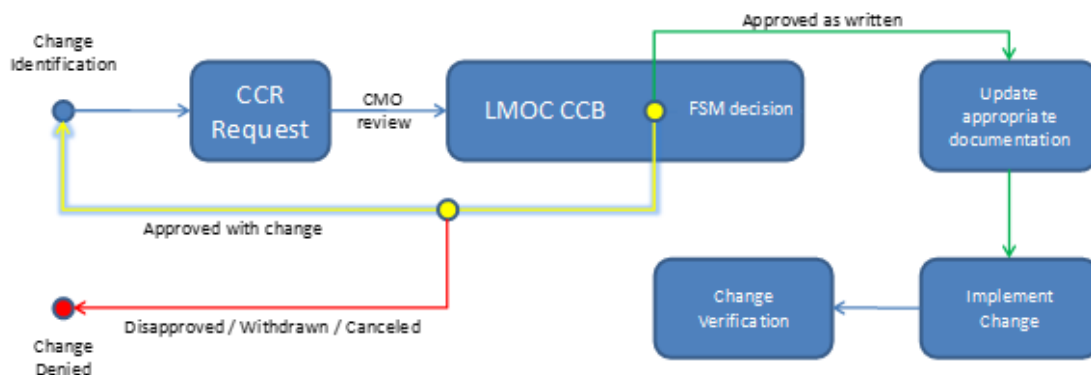


Figure 4-17 Configuration Management Flow Diagram

The general overview of the process is as follows:

1. Any member of the LMO team may recognize the need for a change and submit a Jira ticket to internal review.
2. The internal review will do one of the following:
 - a. Terminate the Jira ticket, by one of the following actions:
 - i. The Jira ticket is disapproved by the internal review
 - ii. The Jira ticket is withdrawn by the original ticket sponsor
 - iii. A previously approved ticket is canceled by the internal review
 - b. Approve the ticket with changes
 - i. The ticket originator will modify the ticket as requested by the internal review and re-submit the ticket into the review status
 - c. Approve the ticket
3. All relevant documentation will be updated.
4. The change will be implemented
5. The change will be tested in a safe environment
6. Test results will be presented to the LMOC O-CCB
7. The change will be installed at the operational level

A critical component of CM is version tracking and management. The LMOC will use the CM repository to support multiuser development and product version/branch control. Products submitted in the CM repository are tracked by a unique ID. Any changes to the product will create a new product ID. Products which pass LMOC CCB will be promoted to the blessed section of the repository for operations implementation.

4.2.18. Regression testing and deploying LMOC builds to operations

Throughout the LMOC development and operational cycle, subsystem applications are expected to be updated and delivered to operations for test and subsystem update. Subsystem application updates may be to address requirements or core functionality as part of the LMOC development and I&T phase, or to fix technical Discrepancy Uncovered Reports (DURs) uncovered during operations or subsystem testing. Subsystem version updates will follow the configuration management plan and processes.

Upon delivery of a subsystem application update to the LMOC, the CCB will open a new subsystem version entry as per the CM plan and capture the new subsystem version, installation manual, and addressed TDRs. A copy of any updates will be entered into the CM repository as part of the mission archive and replicated in the LMOC Reference Image at EROS in Sioux Falls, South Dakota for backup.

After a new subsystem version has been opened by the CCB, applicable operations and administration team members review and decide if documentation and functionality meets expectations. If the delivered release fails to meet the intent of the update or has insufficient documentation, the update will be rejected and the CCB will coordinate with the vendor on an updated release. Accepted releases are opened for formal review and signature by the USGS customer.

After acceptance by the USGS, a configuration change request Jira ticket is opened to install and test the new software version. Once the ticket is approved, the software will be installed in the bLMOC or some other non-operational system. As the software is installed the administration team will capture redlines and note any changes that should be made in the installation procedure for smooth install on operational systems. When install is complete the test system will be released to operations for regression testing and validation of addressed DURs. If issues are found in testing the associated ticket will be updated and discussions will be opened with the vendor to address issues. The ticket is closed when the operations tester is satisfied with updated software performance.

Install of the software on operational systems will occur only after successful install and checkout of the test system. Operational system updates will require a new CCR and are coordinated with operational activities and the operational team.

4.2.19. Data and configuration backups and archive

Function of the LMOC and ongoing operations relies on several different types of data. The LMOC infrastructure supports archive and backup of required operations files, documents, and software as a mitigation against system issues and data loss. LMOC data is captured in the Trending repository, CM repository, controlled repository, and document repository. Repositories are replicated to the bLMOC and offsite at EROS in Sioux Falls South Dakota. LMOC data backed up at EROS from the Landsat Reference Image.

Stored State of Health (housekeeping) data from the observatory is received in the LMOC and injected into the STARS trending system. During injection STARS builds a trend database which contains spacecraft engineering data in a format ready for trending and/or statistical analysis. The STARS also statistically reduces telemetry to generate long-term trending (LTT) data set. The processed telemetry database is replicated to the bLMOC and to EROS as a backup.

Operations of the LMOC also requires strict management of scripts, procs, and other configuration files which are used to configure subsystems and execute automation. Script, proc and configuration files interact and are carefully CM controlled and version tracked within the CM repository which is governed by the CM plan. LMOC software, software patches, and operating system gold images are also included and tracked within the CM repository. The CM

repository is replicated to the bLMOC and to the EROS LMOC reference image as a mission backup.

Operational products (files containing data used by LMOC subsystems, observatory, external systems, etc.) keep automation and routine operations functional. Specified operational products are copied to the LMOC Controlled Repository as an on-site mission archive by Mercury. The Controlled Repository is replicated to the bLMOC and at EROS LMOC reference image as a mission backup.

Operational documents (SOPs, COPs, training documents, etc.) are CM stored within the LMOC document repository. Operational products are controlled as per the CM plan and tracked with unique IDs within the document repository. The document repository is replicated to the bLMOC and to the EROS LMOC Reference image as a mission backup.

In the event of a bLMOC failover, required operational products will be present in the bLMOC Controlled Repository so multi mission operations can be continued without interruption. If the LMOC and the bLMOC become unavailable a replacement operations center can be created from the LMOC reference image stored at EROS.

4.2.20. bLMOC contingency operations

A backup Landsat Mission Operations Center (bLMOC) will provide a safeguard against LMOC system, LMOC network, or LMOC facility issues. The bLMOC is designed to have the major capabilities of the LMOC to allow for continued operations in the event of an LMOC outage – however the bLMOC will have reduced redundancy when compared the LMOC.

The bLMOC will be located in a facility separated from the LMOC so that it can function independently in the event of a critical failure, malicious attack, or issue at the LMOC facility. The location of the bLMOC is TBD. The LMOC will routinely share/update command and control data and information with the bLMOC so that the bLMOC can assume operations of the Landsat mission with no risk to mission health and safety, and minimal impact to image data collection.

The Flight Operations Team Manager can transfer operations responsibility to the backup location at any time. The switchover may be initiated from the backup mission operations center, in the event the primary location is unexpectedly non-operational. To ensure that the backup capability is ready, failover operations to and from the bLMOC will be exercised on a periodic basis.

4.2.21. Deorbit and passivation

At the end of the mission, the Landsat satellites will undergo decommissioning. The decommissioning of the L8 and L9 observatories will comply with the NASA Safety Standard Guidelines and Assessment Procedures for Limiting Orbital Debris (NSS 1740.14) and the NASA Policy for Limiting Orbital Debris Generation (NPR 8715.6). All applicable segments, management paths, and support teams will be notified of the decommissioning plan and timeline 90 days prior to execution.

At the end of mission a series of de-orbit burns will be executed to lower the orbit and do a controlled reentry. The burns will lower perigee in increments of 50 km. There will also be an additional burn at the beginning of the burn sequence to lower apogee in order to move the Landsat satellite below the 705-km altitude orbit where the afternoon and morning constellation spacecraft operate. The final burn will target a perigee altitude of 50 km and be timed such that the reentry occurs in a location that is compliant with NPR 8715.6. The spacecraft will maintain attitude control for all propulsive maneuvers including the last disposal maneuver.

4.3. Misuse cases:

Several use misuse or contingency scenarios are possible during LMOC operations. The most likely or critical scenarios need to be understood and developed, however full contingency testing may be very disruptive or risky. Identification of misuse cases is an important exercise for identifying and mitigating key operational risks with a robust system design.

4.3.1. IT security incident response

While the LMOC is designed and configured to meet NASA and USGS IT security standards, it is possible that an IT security incident may occur. In the event of an IT security incident suspected and confirmed incidents must be reported immediately to:

- NASA Security Operations Center (SOC)
- EROS Information Security Manager (ISM)
- USGS Computer Security Incident Response Team (CSIRT)
- EROS LMOC Agency Security Manager (ASM)
- LMO IT Security Lead

Upon notification, the LMO IT Security Lead will gather information from any/all personnel involved and fill out and submit the appropriate forms for NASA and USGS.

Upon notification the NASA SOC is responsible for notifying the local GSFC IRM and the Deputy GSFC IT Security Manager (D-ITSM) for Operations, who will contact the appropriate parties to initiate an investigation. The IRT typically contact the appropriate organization Information Technology Managers (IT Managers), security engineers and system administrators to begin the investigative process. The team may additionally contact the Office of the Inspector General and the Personnel Security Division, depending on the nature of the incident.

The EROS LMOC ASM is responsible for notifying any EROS/USGS personnel not listed above who may need to be informed of the incident.

In the event of a breach, LMOC personnel will take no action, other than reporting/notification. GSFC IRT will conduct the investigation.

4.3.2. Hardware failures

Hardware failures are expected over the LMOC design life and the ANS supports monitoring of infrastructure hardware and functions. If a failure is detected the system administration team will be alerted and coordinate to verify the scope or nature of the issue. If an actual hardware fault is in progress the system administration team will notify operations. The administration team may require the operations team to test or verify various LMOC functions to understand the full scope of the problem.

If the fault is isolated to a subsystem the administration team will coordinate with operations on a hardware failover to a redundant string. If the fault impacts multiple LMOC subsystems, users, or key interfaces, the issue will be escalated as per the anomaly response plan. Operations can fail over to the bLMOC as per the discretion of the FOT manager and the USGS FSM. The administration will work to fix the hardware fault by either replacing the component, invoking a support agreement, or onsite troubleshooting.

4.3.3. Disaster recovery scenarios

A disaster scenario assumes complete loss of either the LMOC, bLMOC, or both the LMOC and bLMOC. Assuming one of the control centers is still functional, the operations team will initiate failover procedures as needed WHEN IT IS SAFE FOR THE TEAM TO DO SO. A disaster scenario may prevent safe travel or interrupt ops team coordination so there may be a delay in operational response until a clear picture of the disaster scope is understood. The operations team will keep the local USGS FSM and EROS center apprised of status and disaster impact as communications allow.

Assuming either the primary or backup control center is undamaged and powered, the operations team will first attempt to restore two way communications with the observatories to ensure health and safety at the remaining functional site. If adequate operations team, network, and external partner support is available, nominal operations will resume and operations will begin the effort to triage and restore operations at the impacted control center if possible.

It may be possible that a widespread disaster has disabled operations at both the LMOC and the bLMOC – or the operations team is unable to travel to reach and staff a control center for an extended period of time. In this scenario it is likely the onboard command load will expire and nominal observatory operations will cease. The end of a command load contains commands to

assert ACS Nadir mode, cease instrument recording, and safe the instruments. If no ground command is received for 3 days, onboard logic will reset the spacecraft receiver and begins a timed failover process to the spacecraft B side. Regardless of the spacecraft's ability to safe itself, the operations team will strive to maintain nominal operations and a two way link with each spacecraft.

Damaged or destroyed control centers will need to be restored by the Landsat team. If the control centers are damaged beyond repair, the USGS may invoke the plan to utilize the LMOC reference image stored at DPAS and to create a new replacement operations center. Hardware will be procured as quickly as possible and the control center will be rebuilt in a manner to prioritize restoration of command and telemetry to the observatories. (If necessary GSFC Flight Dynamics Facility can be temporarily be used to generate orbit determination products and predicts to determine station in-view times.)

The LMOC reference image contains the OS gold images to recreate LMOC subsystem environments. After a subsystem template is created, the subsystem software can be installed and configuration recreated from the CM repository. Operational products and operational documents required to resume operations can be accessed from the LMOC reference image as well.

After a system is restored, the operations team will perform a quick functionality check and then promote it to operations. Since systems will be phased in one at a time, manual operations and 24/7 staffing may be required until enough systems have been restored to resume automation. When restoring an operations center priority will be placed on restoring one of each system before starting restoration of redundant or backup systems.

4.4. Functional Threads

The LMOC operations are illustrated by a series of functional threads. These threads provide an overview of major concepts and illustrate step by step how important activities are accomplished. The set of functional threads help in understanding how the LMOC system behaves, and serves as a tool to aid in verifying the completeness of the Operations Concept. The scenarios do not portray every possible situation that may be encountered during operations but serve as a basis for further operations planning. Lesser steps will be omitted to distil a simple high level process to illustrate the core concept of the functional thread.

Each scenario includes a brief description, a list of assumptions, a data flow figure, and a table that describes the key functions and data flows. Each functional thread figure has numerated steps which are explained in the description table.

4.4.1. Daily Activities

The following functional threads occur support daily LMOC operations as part of the nominal operations concept.

4.4.1.1. Command Scheduling

Description: Command loads are generated and uplinked to the observatory. Command loads contain stored commands for spacecraft transmitter management, science data collection, instrument calibrations, instrument management, spacecraft subsystem management, and data playbacks. ROC automation also issues commands for Mission Data Management and real-time operators may issues commands not within the load for periodic observatory housekeeping activities. The same sequence is performed separately for each observatory.

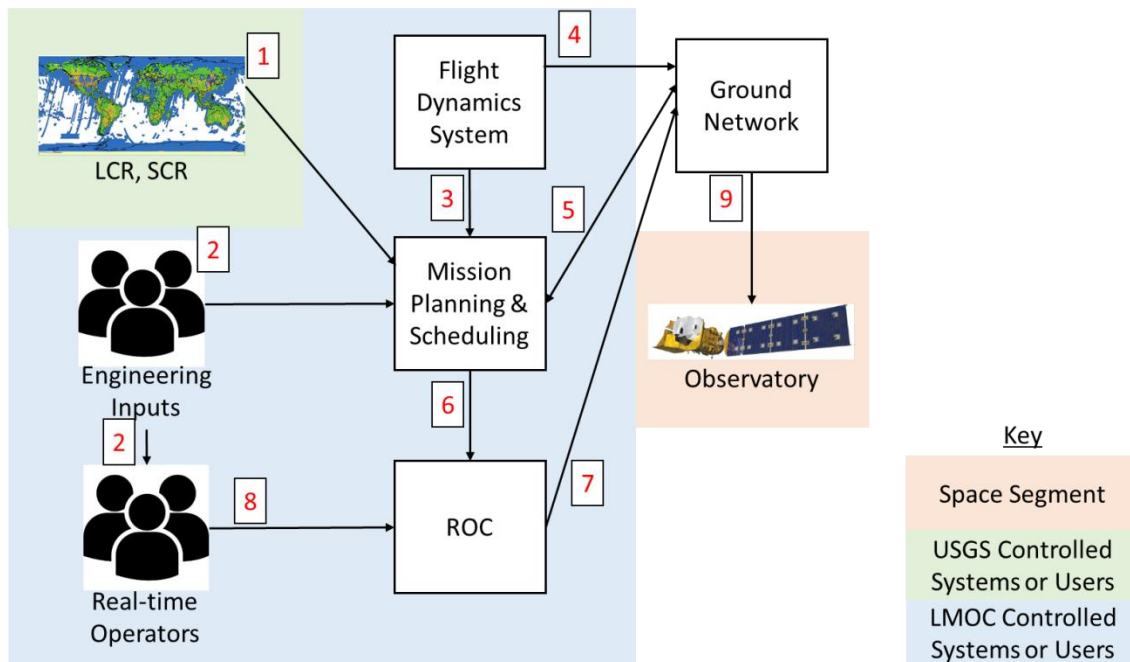


Figure 4-18 Command Scheduling

Step	Description
1	Data collection requests (LCR, SCRs, special calibrations) are submitted from the USGS Data Acquisition Manager and Cal/Val team to the LMOC Mission Planning and Scheduling (MPS) System. The Calibration Opportunities Scheduling Tool (COST), internal to MPS, generates and submits CalCRs to MPS.
2	Observatory engineers within the LMOC submit plans for any upcoming engineering activities to MPS that need to be contained in the stored command load. Commands that need to be issued in real-time during a pass are coordinated with real-time operators.
3	FDS delivers predictive ephemerides to mission planning. Predictive ephemerides are used to schedule ground station contacts when the observatory is in view.

Step	Description
4	FDS delivers observatory acquisition data to the ground stations for pointing and tracking the observatory during scheduled ground contacts.
5	MPS will use the data collection requests and engineering inputs to create a Contact Forecast Request which contains the desired ground contacts to meet upcoming operational needs. Stations will review the requests and provide a Contact Forecast Confirmation with the contacts they can support. The Mission Planning team will release a Confirmed Contact Schedule for the upcoming week once the required number of contacts has been secured.
6	MPS will generate a Master Activity List that Galaxy uses to create a command load that is reviewed, approved, and delivered to the ROC for uplink
7	Two way communication is established with the spacecraft during a scheduled ground contact. Issued commands are encrypted and transmitted to the ground station for uplink. Ground automation on the ROC performs nominal pass activities
8	Real-time operators will issue housekeeping commands and commands submitted by the observatory engineers that are not contained within the spacecraft load or issued by ground automation. Approved and ready command loads are uplinked at this time.
9	Commands are radiated from the ground station via S-band uplink and received by the observatory for execution.

Table 4-1 Command Scheduling

4.4.1.2. Data Downlink and Processing

Descriptions: During a GNE or LGN scheduled contact the observatory downlinks stored and real-time data to the ground station. Received mission data is converted into science products for the user community, stored housekeeping data is converted into trend products for offline engineers, and real-time data is viewed by real-time operations and engineering staff. Observatories may be in contact with different ground stations at the same time and DPAS and the LMOC systems are designed to process simultaneous observatory data streams.

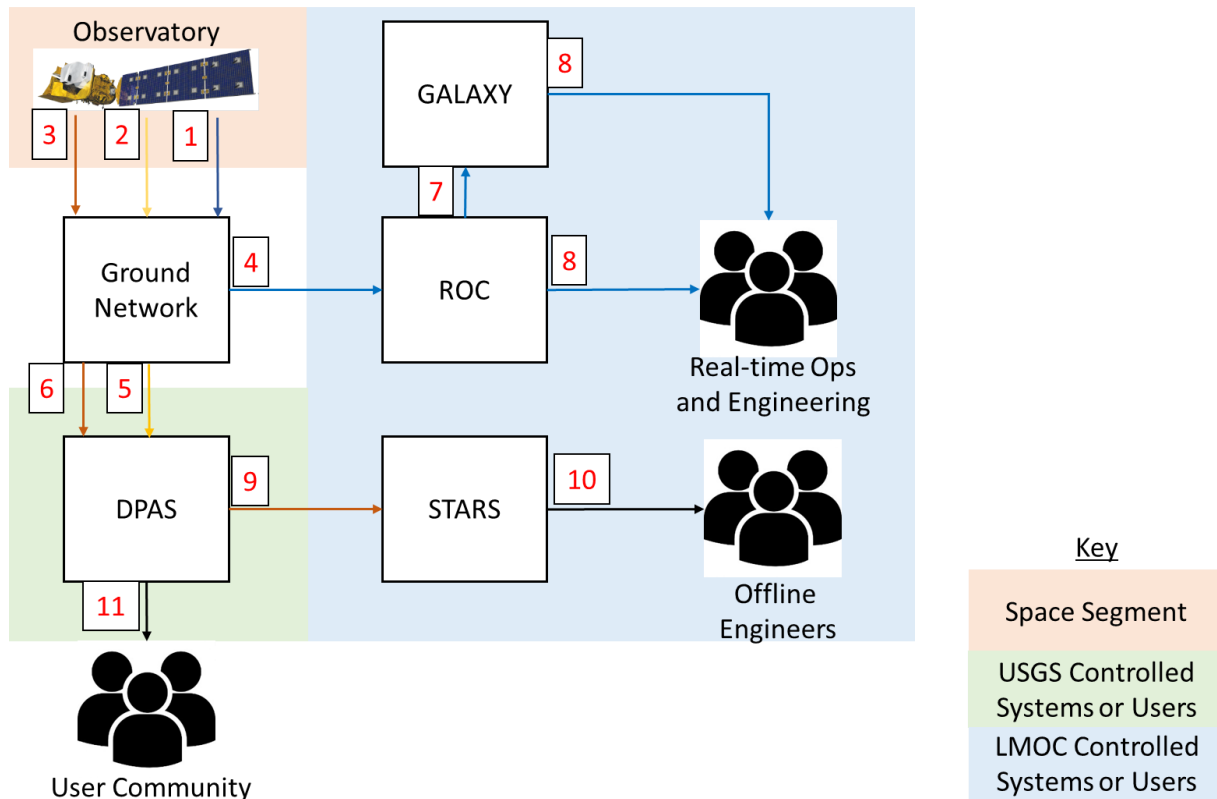


Figure 4-19 Data Downlink & Processing

Step	Description
1	An S-band downlink is initiated with the ground station during a scheduled support. Real-time engineering and housekeeping data is downlinked over the S-band link.
2	.An X-band downlink is initiated with the ground station during a scheduled support. Real-time image/science data is downlinked over the X-band link. If no real-time image/science data is concurrent with the time of the ground support, real-time image/science data will not be downlinked.
3	An X-band playback is initiated by observatory stored command to downlink mission data and SSOH stored on the SSR. Data retransmits are also issued by ground automation and downlinked as an X-band playback.
4	Real-time engineering data is bent-piped by the ground network and delivered to the ROC
5, 6	Real-time image/science data from the observatory is captured at the ground station and transferred to DPAS after the support
6	Playback back-orbit mission, SSOH, and retransmit data from the observatory SSR are captured at the ground station and transferred to DPAS after the support.
7	Real-time engineering data received at the ROC is distributed to all configured GALAXY workstations
8	LMOC real-time operators and engineers monitor real-time data to verify observatory status and health
9	SSOH data received at DPAS is sent to the LMOC for ingest in the STARS trending and archive system.

Step	Description
10	Limit violation reports, command history reports, spacecraft event reports, and engineering plots generated by STARS are used by offline engineers to verify observatory performance and state of health.
11	Mission data received by DPAS is processed and hosted for the user community

Table 4-2 Data Downlink & Processing

4.4.1.3. Mission Data Management

Description: Mission Data Management (MDM) is the process used to schedule, record, downlink, and clear data from the recorder. MDM is a systems concept which relies on mission planning, onboard stored commands, real-time ground automation, station file reports, LFAT, and various LMOC products. L8 and L9 both utilize the MDM process, however each satellite's MDM cycle and MDM products are handled independently within the LMOC.

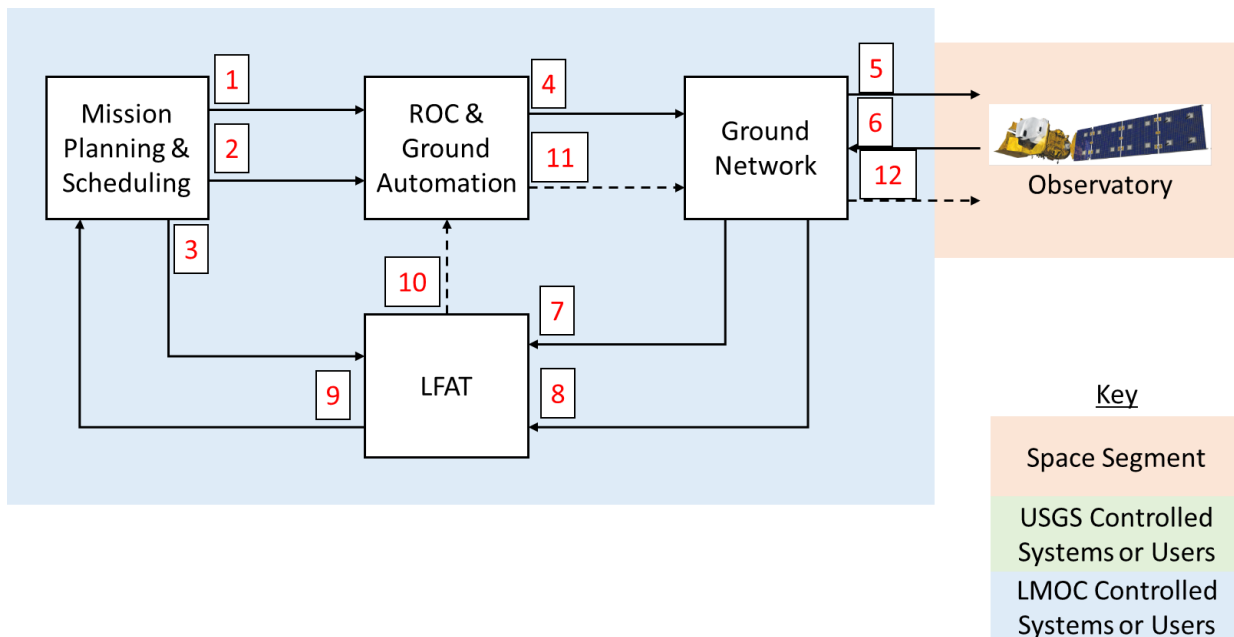


Figure 4-20 Mission Data Management

Step	Description
1	As part of a command load package MPS produces an observatory stored command load which is delivered to the ROC for uplink
2	As part of a command load package MPS produces MDF file which is delivered to the ROC for ground automation timing
3	As part of a command load package MPS produces a SIMT which is delivered to LFAT to map scenes scheduled for imaging during the command load to their mapping to root file ID/interval
4, 5	Prior to the command load start the observatory stored command load is sent from the ROC to the ground station and uplinked to the observatory during a ground contact
NOTE	The observatory stored command load, MDF, and SIMT must become active onboard the observatory, on the ROC, and within LFAT when the stored command load is scheduled to start

Step	Description
6,7	During a ground contact a SSR directory listing is dumped from the observatory and sent from the ground station to the LFAT in real-time. The directory listing contains the status of files on the observatory SSR
8	After the ground contact completes, the ground station generates a GPR which is sent to LFAT. The GPR contains the list of files received during the ground contact and completion status
9	LFAT compares the GPR, SSR Directory Listing, and the SIMT file and determines which SSR root files are available for use. LFAT writes the information to the SIFMT file which is delivered to MPS for use in building the next command load.
10	LFAT compares the GPR, SSR Directory Listing, and the SIMT file and generates an unprotect and delete list for all files which have been confirmed as received on the ground and complete. The unprotect and delete list will be used by ground automation to generate unprotect and delete commands for the next uplink opportunity
11,12	On the next uplink opportunity the unprotect and delete commands are sent from the ROC to the ground station and then uplinked to the observatory for execution

Table 4-3 Mission Data Management

4.4.2. On Orbit Maneuver Planning

Propulsive maneuvers must be executed by the observatory to avoid space debris and to maintain the proper orbit and assigned constellation control box. Orbit maintenance maneuvers can be predicted and planned well in advance of execution by FDS. Risk mitigation maneuvers are often planned and executed with little prior warning to avoid conjunctions. FDS works closely with the Conjunction Assessment team to identify conjunction risks and to verify planned maneuvers are safe to execute. All maneuvers are reviewed by the by operations and the USGS FSM. Approved maneuvers are uplinked and executed by the observatory. The L8 and L9 observatory orbits will be maintained and managed separately by operations.

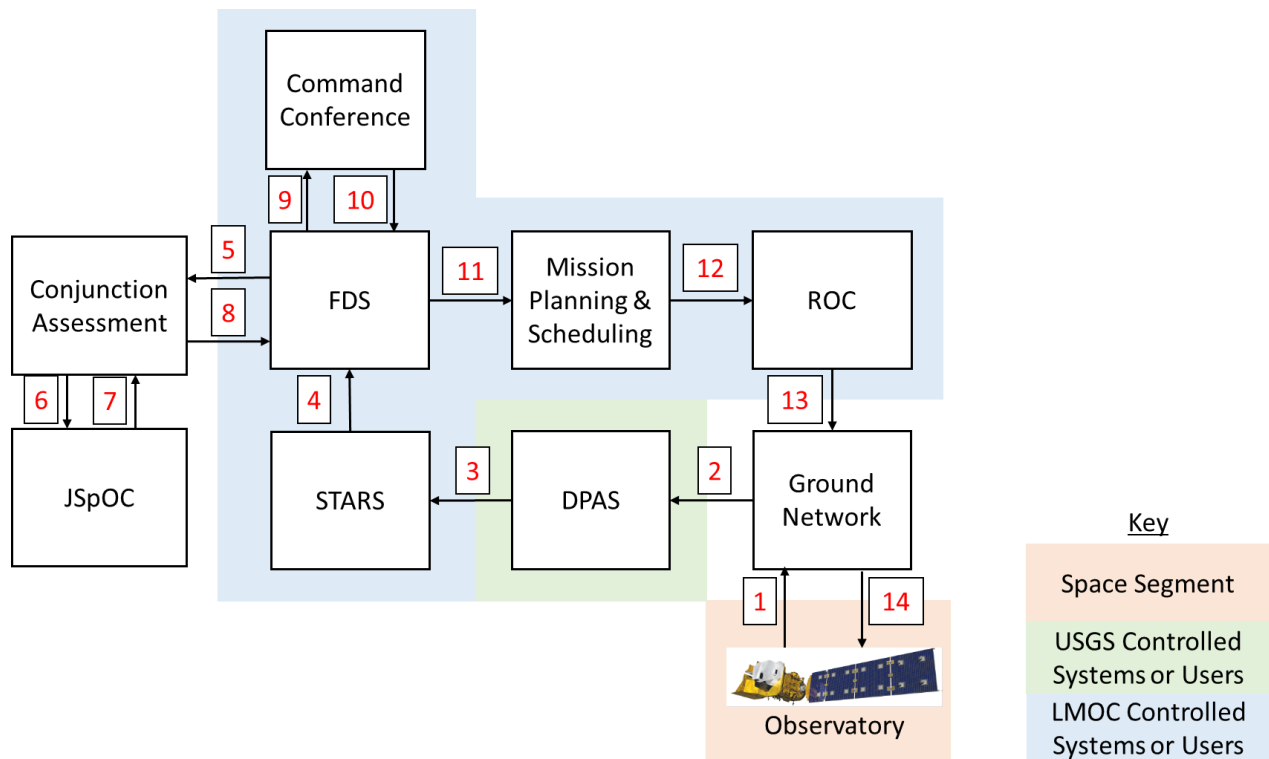


Figure 4-21 Maneuver Planning

Step	Description
1	SSOH data is downlinked from the observatory to the ground station during a ground contact
2	After the ground contact the SSOH data is transferred to DPAS for archive and distribution
3	DPAS sends the SSOH data to STARS where it is ingested and generates engineering data products
4	Engineering data products which contain position and attitude data are delivered to FDS to use in orbit determination and observatory ephemeris updates.
5,6	Observatory ephemeris updates are delivered to the conjunction assessment team which delivers them to JSpOC
7	JSpOC uses the observatory ephemerides to screen against cataloged and tracked space objects. Objects which may pose a threat compiled along with close approach times and tracking details and delivered to the Conjunction Assessment team.
8	The conjunction assessment team reviews the JSpOC information and evaluates conjunction risks. Risks are compiled in a report for FDS review
NOTE	If a conjunction looks unacceptable or a nominal orbit maintenance maneuver is needed, FDS will generate a maneuver plan. A maneuver plan ephemeris is screened by the Conjunction Assessment team to ensure the maneuver does not introduce new conjunction threads or mitigates existing conjunction threats if applicable
9	Proposed maneuver plans are submitted to a USGS led command conference. Maneuver plans and parameters will be reviewed by operations and the USGS FSM
10	If the maneuver is approved FDS is notified and allowed to proceed with maneuver delivery

Step	Description
11	FDS delivers maneuver products to MPS which incorporates maneuver details into the command load
12	The command load is reviewed, approved and delivered to the ROC
13, 14	On a ground contact prior to the maneuver the stored command load is sent to the ground station and uplinked to the observatory

Table 4-4 Maneuver Planning

4.4.3. Data Archive

The LMOC utilizes and generates a wide array of data products, software, and documentation. Data must be CM controlled and properly archived within the LMOC for operational access and use. To mitigate LMOC failure scenarios, LMOC data is replicated to the bLMOC so it is available for operation center failover. LMOC data is also replicated to the LMOC Reference Image at the USGS EROS facility. The LMOC Reference Image is a nonoperational backup of LMOC data which can be used to rebuild an operations center in the event of a disaster recovery scenario.

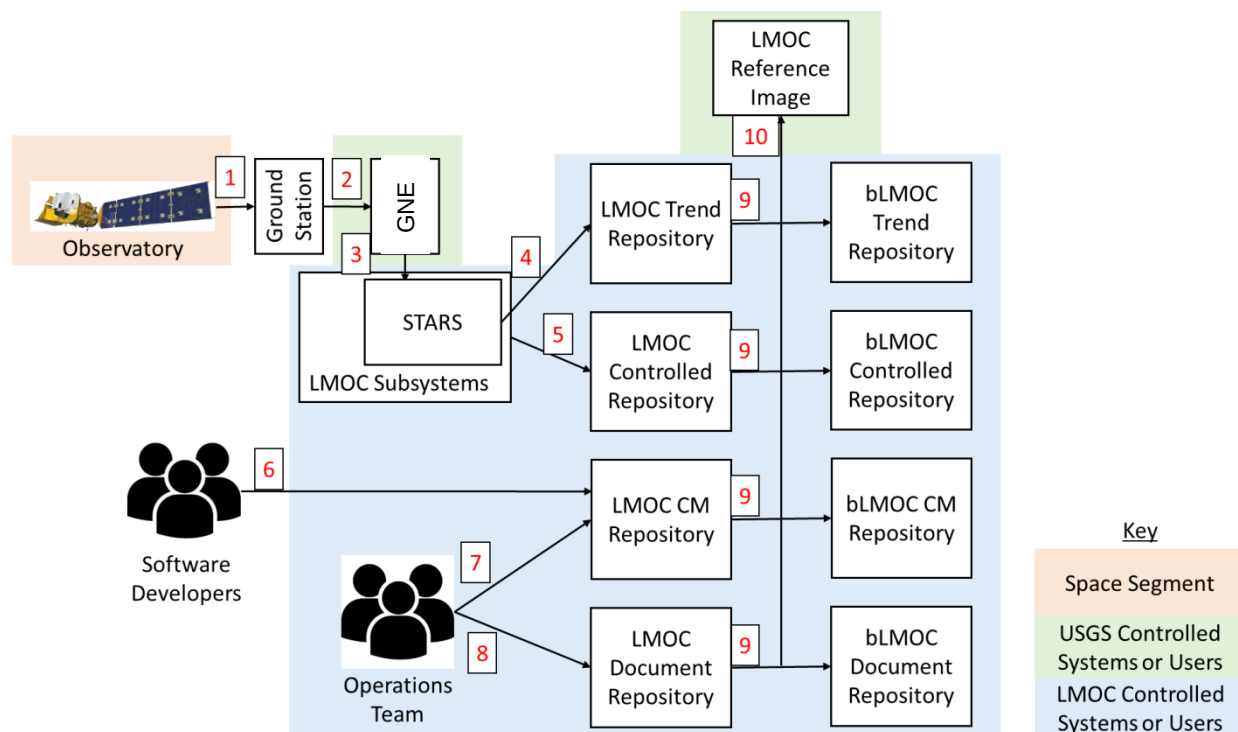


Figure 4-22 Data Archive

Step	Description
1, 2	SSOH data is downlinked from the observatory to the ground station and is transferred to DPAS after the ground contact

Step	Description
3	DPAS archives the SSOH data locally and delivers the SSOH data to STARS for processing. SSOH data is injected into STARS.
4	STARS statistically reduces telemetry to generate long-term trending (LTT) and other engineering products. LTT and other pertinent products are archived in the LMOC Trend Repository
5	Operational data products delivered to the LMOC or generated by LMOC systems are captured by Mercury and archived in the LMOC Controlled Repository
6	Accepted software vendor software releases, patches, and documentation from software developers are archived in the LMOC CM Repository
7	Scripts, configuration files, and OS gold images developed or managed by the operations team are CM controlled and archived in the LMOC CM Repository
8	Operations SOPs, COPs, and documentation is CM controlled and archived in the LMOC Document Repository
9	LMOC repositories are replicated to the bLMOC
10	Periodically LMOC repositories are also replicated to the LMOC Reference Image at USGS EROS as an information backup

Table 4-5 Data Archive

5. Maintenance Operations

5.1. System update concept

System patching, security scanning, and testing of software updates are supported largely through the use of an on-site Integration & Test (I&T) environment. Any changes of these three types is fielded first in the I&T environment before being submitted to OCCB for deployment in the operations environment.

The majority of software packages are managed through the use of Microsoft Windows Server Update Services (WSUS) and Red Hat Satellite Server (SatServer). Upon creation of a CCR, these infrastructure tools are used to fetch new updates via the internet, then apply them to selected virtual machines (VM's). The VM's in the I&T environment are updated first, and tested by the FOT GSE using the automated test frameworks. Following checkout by the GSE, the CCR is updated with the pertinent information regarding software update steps, testing steps, reboot requirements, etc. The CCR is then promoted to OCCB for approval before applying the updates to the operational environments.

The approach to monthly patching and scanning utilizes the processes outlined above, but take place on a regular monthly cadence. The target timeline, shown in Figure 5-1, is designed to complete all monthly activities with time to spare, allowing delays resulting from critical or time-sensitive satellite operations.

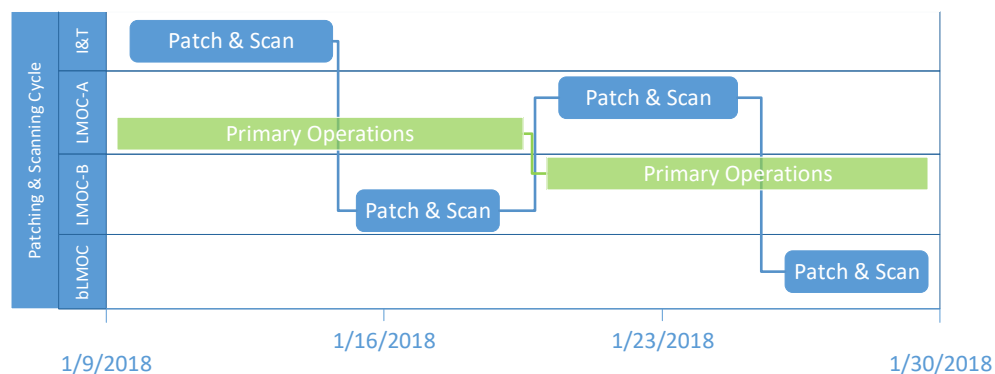


Figure 5-1 Monthly IT Patching & Scanning

6. Acronym List

Acronym	Definition
ACK	Acknowledge
ACS	Attitude Control System
ANS	Alerts and Notifications System
AOS	Acquisition Of Signal
API	Application Programming Interface
ART	Anomaly Resolution Team
ASM	Agency Security Manager
ATS	Absolute Time Sequence
AWT	Attitude Window Table
BLMOC	Backup Landsat Mission Operations Center
CARA	Conjunction Assessment Risk Analysis
CCB	Configuration Control Board
CCR	Configuration Change Request
CCS	Confirmed Contact Schedule
CCSDS	Consultative Committee for Space Data Systems
CDMP	Configuration & Data Management Plan
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CFDP	CCSDS File Delivery Protocol
COST	Calibration Opportunities Scheduling Tool
COTS	Commercial Off-the-Shelf
CPAW	Collection Planning and Analysis Workstation
CPU	Central Processing Unit
CSIRT	Computer Security Incident Response Team
CTP	Command and Telemetry Processor
DAM	Data Acquisition Manager
DCRS	Data Collection & Routing Subsystem
DID	Data Item Description
DITL	Day In The Life
DMS	Data Management System
DMU	Drag Make Up (maneuver)
DPAS	Data Processing and Archive System
DTM	Daily Telemetry Message
EGSE	Electrical Ground Support Equipment
EPS	Electrical Power Subsystem
EROS	Earth Resources Observation and Science
ESC	Earth Science Constellation
ESD	Electrical Static Discharge
ESMO	Earth Science Mission Operations
FAT	Factory Acceptance Test
FDF	Flight Dynamics Facility

FDS	Flight Dynamics System
FOM	Flight Operations Manager
FOR	Flight Operations Review
FOSS	Free and Open Source Software
FOT	Flight Operations Team
FSL	Flight Systems Lead
FSM	Flight Systems Manager
FST	Flight Support Team
FSW	Flight Software
GDMS	General Dynamics Mission Systems
GFE	Government Furnished Equipment
GFP	Government Furnished Property
GNE	Ground Network Element
GORTS	Ground Ops Readiness Test
GOTS	Government Off-the-Shelf Software
GPR	Ground Post-pass Report
GPS	Global Positioning System
GRT	Ground Readiness Tests
GSCCB	Ground System Configuration Control Board
GSE	Ground System Engineer
GSFC	Goddard Space Flight Center
GSI	Ground System Infrastructure
HSSDB	High Speed Serial Data Bus
IDSR	Image Data Schedule Report
INC	Inclination
IRM	Incident Response Manager
IRT	Incident Response Team
ISM	Information Security Manager
ITSEC	Information Technology Security
ITSM	IT Security Manager
IWT	Instrument Window Table
JASON	Joint Activity Simulation for Observatory and Network
JSPOC	Joint Space Operations Center
KSC	Kennedy Space Center
LCR	LTAP Collection Request
LDCM	Landsat Data Continuity Mission
LEO	Low Earth Orbit
LFAT	Landsat File Accounting Tool
LGN	Landsat Ground Network
LLSRD	Lower Level Systems Requirements Document
LMER	LMOC Mission Equipment Room
LMO	Landsat Mission Operations
LMOC	Landsat Multi-Satellite Operations Center

LMOCDRB	LMOC Discrepancy Report Board
LMOCDR	LMOC Requirements Document
LOS	Loss Of Signal
LSDS	Landsat Satellites Data System
LSE	Lead Systems Engineer
LSIMSS	Landsat Scalable Integrated Multi-Mission Support System
LSR	Launch Support Room
LTT	Long Term Trending
MAL	Master Activity List
MDF	MOC Directive File
MDM	Mission Data Management
MLT	Mean Local Time
MOC	Mission Operations Center
MORM	Mission Operations Readiness Manager
MORT	Mission Operations Readiness Team
MOS	Mission Operations Simulation
MOVE	Mission Operations Voice Enhancement
MPS	Mission Planning System
MRT	Mission Readiness Test
MTASS	Multi-mission Three-Axis Stabilized Attitude Support System
MTR	Mission Transition Review
NASA	National Aeronautics and Space Administration
NEN	Near Earth Network
NFS	Network File Share
NISN	NASA Integrated Services Network
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NOOP	No Operation
NPR	NASA Procedural Directive
NSS	NASA Safety Standard
OBC	On Board Computer
OCCB	Ops Configuration Control Board
OEM	Original Equipment Manufacturer
OLI	Operational Land Imager
OMB	Office of Management and Budget
OOL	Out Of Limits
OPS	Operations
OTS	Off The Shelf
PDB	Project Database
PDR	Preliminary Design Review
PIE	Payload Interface Electronics
PLAR	Post Launch Assessment Review
PLN	Plan

PMO	Program Management Office
POA&M	Plan of Actions and Milestones
PSR	Pre-Ship Review
PTR	Post Test Review
QASP	Quality Assurance Surveillance Plan
RFR	Run For Record
RMM	Risk Mitigation Maneuver
ROC	Real-time Operations Console
ROS	Relative Operations Sequence
RSPP	Robotic Systems Protection Program
RVM	Requirements Verification Matrix
SAT	Site Acceptance Test
SCF	Scheduler Contact File
SCS	Stored Command Sequence
SDVF	Software Development & Validation Facility
SEMP	Systems Engineering Management Plan
SIFMT	Scene Interval File Mapping Table
SIMT	Scene Interval Mapping Table
SMS	Short Message Service
SOC	Security Operations Center
SOS	Spacecraft / Observatory Simulator
SPV&VP&M	System Performance Verification And Validation Plan And Matrix
SSOH	Stored State Of Health
SSR	Solid State Recorder
STK	Systems Tool Kit
STOL	Space Test & Operations Language
SWOTS	Shrink-Wrap OTS software
TBD	To Be Determined
TCS	Thermal Control System
TDR	Test Discrepancy Report
TDRSS	Tracking Data Relay Satellite System
TIRS	Thermal InfraRed Spectrometer
TRR	Test Readiness Review
TRT	Transition Readiness Test
USAF	United States Air Force
USGS	United States Geological Survey
USSTRATCOM	US Strategic Command
UTC	Universal Time Coordinated
VDD	Version Description Document
WAN	Wide Area Network
WDC	Wideband Data Collector
WRS	Worldwide Reference System
WSUS	Windows Server Update Service